The Locus of Qualia

Contextual Semantic Binding, Experiential Architectonics, and a Research Program for Realization Mapping

Not a mint-shaped object hidden in matter, but a constrained way an encounter becomes experience.

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Abstract

What would it mean to specify the taste of mint mathematically? This paper proposes a contextual binding event as the explanatory unit for qualitative specificity: an organized occurrence in which sensory differentiation, bodily state, temporal history, attention, and other relevant constraints jointly determine a particular experiential character. The proposal extends the Semantic Binding Principle and its distinctions among identity, support, constitution, and realization. It does not follow deductively from that semantic framework. The qualitative event, its hypothesized constitutive organization, its physical realization, and the observations used to investigate it remain separately typed.

Mint provides a worked domain: weak and strong mint, peppermint and spearmint, fresh and dried preparations, decay-associated qualities, and evolving aftertaste. These form candidate regions and trajectories in a context-indexed experiential atlas, not predetermined coordinates in a universal Euclidean space. Typed dimensional distinction, statistical independence, geometric orthogonality, and causal selectivity are distinguished. Experiential-content coordinates are also separated from the Ableman Consciousness Ontology's binding-regime axes, Semantic Manifold Competence's operational capacities, and the epistemic properties of a research account.

A source-integrated architecture assigns different responsibilities to Locus, Global Term Key, Dynatology, Semantic Manifold Topology, Semantic Binding Quality, and the GMB/crystallization lineage as governance of the research account. Dynatology contributes explicit reachability; SMT contributes declared continuity auditing rather than an assumed geometry of experience; SBQ supplies candidate criteria for model integrity rather than a consciousness score. A new path-realization requirement shows why individually realizable projected transitions may form an unrealizable sequence when a chart identifies states with different continuations. A finite-path lifting criterion and a worked counterexample make

this issue testable.

The contribution is a falsifiable research architecture, not an empirical demonstration of phenomenal constitution. Proposed studies compare typed, contextual, history-sensitive binding models with strong alternatives on held-out observations, interventions, and sequences. Limited mathematical results and two executable synthetic fixtures establish only properties of stipulated models. The strongest hypothesis remains that qualitative specificity is constituted by a particular realized binding organization; prediction, operational competence, governance admission, and causal correspondence do not individually establish that hypothesis.

Keywords: qualia; semantic binding; phenomenal structure; contextual realization; architectonics; experiential atlas; causal abstraction; psychophysics; representational geometry; mint; consciousness; identifiability.

1. Introduction: From the Object for Mint to the Constitution of an Experience

The expression *the taste of mint* appears to identify one familiar quality. On examination, it covers many distinguishable encounters. A faint aftertaste differs from an intense initial sensation. Peppermint and spearmint may be discriminated without being classified as entirely unrelated. A fresh leaf, a dried preparation, and a decay-associated odor can preserve some recognizably mint-related character while differing on other dimensions. An apparently identical stimulus can also be encountered under different histories, expectations, and bodily conditions.

These examples motivate the present hypothesis; they are not a dataset. Their significance is that the explanatory target may be a structured family of contextual realizations rather than an isolated, context-free primitive called `MINT`.

The immediate point of departure is an interview transcript supplied by the author. Donald Hoffman challenges computational accounts to specify a mathematical structure for a particular conscious quality and explain why that structure is mint rather than chocolate. He distinguishes the absence of an explanatory mapping from the computational expense of evaluating an already specified theory [T1, approximately 19:59-24:30]. The distinction is useful, but the question admits several interpretations. A context-free matrix is one possible target; a contextually embedded causal organization is another. A computational theory need not restrict itself to the former.

The present proposal therefore does not declare Hoffman's challenge refuted. It asks whether a better unit of analysis can make part of the challenge experimentally tractable:

Which realized binding organization specifies this experience,

and which transformations of its physical support predict its changes?

The proposal has a deliberately strong conceptual ambition. Qualitative character may be constitutively relational: not merely influenced by context after a complete quale has already formed, but individuated through the organization of a situated event. Its empirical ambition is narrower and more immediate. A binding-sensitive model should explain and predict discriminable changes better than competing models under controlled conditions.

This paper develops those ambitions separately. It does not equate having a useful semantic model with having a theory of phenomenal existence.

1.1 Contributions

The paper makes six proposed contributions. It identifies a contextual binding event as a candidate explanatory unit for qualitative specificity; separates identity, support, constitution, realization, and evidence; develops local, potentially mixed discrete-continuous descriptions without presupposing independent sensory axes; distinguishes content from system-regime and competence descriptions; defines both intervention-correspondence and complete-path realization tests; and provides an experimental program in which the resulting structure must earn its role against strong alternatives. The source integration assigns responsibilities without treating related author manuscripts as independent empirical corroboration.

The novelty claimed is this particular integration and its explicit research contracts. The paper does not claim to originate relational accounts of experience, multidimensional psychophysics, contextual perception, representational geometry, or causal abstraction.

1.2 Source discipline

Four kinds of material are kept distinct. The supplied author manuscripts provide the conceptual lineage and explicitly bounded source commitments [M1-M2, M4-M15]. The earlier draft also cited a Semantic Field Mechanics brief [M3]; that bibliographic lineage is retained, but that separate brief was not newly supplied or re-examined for this revision. The interview provides Hoffman's stated challenge, not proof of his metaphysics or a verified account of the entire consciousness literature [T1]. Published sources provide the prior draft's selected empirical and methodological context [R1-R12]. The literature register is retained rather than presented as a new systematic literature verification. It was not externally revalidated during this closed-corpus pass. New definitions, applications, counterexamples, and tests are identified as constructions of this manuscript. Source discrepancies are recorded in Appendix E rather than silently amended.

The literature discussion is selective rather than systematic. No priority claim or exhaustive novelty search is asserted.

2. Four Questions That Must Not Be Collapsed

The phrase *define a quale* can refer to at least four tasks.

Descriptive specification identifies the differences that characterize an experience: cooling, aromatic character, intensity, temporal profile, and so on. **Predictive specification** maps independently measured conditions onto reports or other experience-related observations. **Constitutive specification** identifies what organization makes an event the particular experience it is. **Existential explanation** explains why such organization is experienced at all.

These tasks constrain one another but are not interchangeable. A good predictor can leave constitution underdetermined. A detailed description can omit physical realization. A proposed identity between organization and experience can remain empirically ungrounded. The distinction between explaining functions and explaining phenomenal experience is central to Chalmers's formulation of the explanatory problem [R1].

This paper primarily develops descriptive and predictive specification, and formulates a constitutive hypothesis. It does not assume that success at the first two levels settles the fourth.

2.1 Semantic, qualitative, and lexical distinctions

A lexical category is a conventional label such as *peppermint*. A semantic binding is a structured, consequential relation within a specified context. A qualitative event is the occurrence of experience whose character is at issue. None is definitionally identical to the others.

The term *semantic* is used here in the framework's broader sense of differential, role-sensitive, constraint-governed organization, not merely as linguistic meaning. Nevertheless, broadening the term cannot do the empirical work. A circuit that discriminates inputs is not thereby shown to feel them. A language model's elaborate description of mint is not itself evidence that it tastes mint. Conversely, lacking the word *mint* does not disqualify a subject from having a mint-related experience.

The research program must therefore remain open to prelinguistic experience, unfamiliar sensations, and experience without explicit conceptual recognition. It must also distinguish a change in what an experience is called from a change in how it feels.

2.2 Artifact does not mean fabrication

The initiating intuition describes a quale as a *contextual artifact at its fundamental representation*. Here, *artifact* means a particular constituted product or

event, not something unreal, deceptive, or necessarily manufactured by a human.

Three objects must be separated:

$$\begin{aligned} &\text{specification of an experience-related structure} \\ &\neq \text{a realization of that structure} \\ &\neq \text{a record describing the realization} \end{aligned} \quad .$$

The strong hypothesis concerns the organization of the experiential event. The research implementation manipulates records and models of that organization. Saving a JSON object with a `cooling` field does not instantiate cooling experience.

3. Conceptual Lineage and Related Research

3.1 Locus, binding, and architectonics

The author's *The Locus of Meaning* treats meaning as constituted through structured binding rather than located exclusively in an object or an interpreter. It distinguishes current realization support, standing structures that preserve possibilities for reactivation, and constitutional invariants across admissible realizations. It also cautions that a semantic domain is not automatically a mathematical manifold [M1, Sections 6-7].

The manuscript stored as *What Locus of Meaning Adds.md* explicitly distinguishes semantic binding, recursive meta-binding, and reflexive meta-binding. Bindings have typed roles, constraints, provenance, context or jurisdiction, and consequences; a binding may itself become an element in a higher-order binding [M2, Abstract]. These are conceptual commitments, not evidence that any particular phenomenal event has been explained.

The later *Semantic Field Mechanics* brief emphasizes transformation laws and invariants, treats semantic architectonics as organization built from bindings, and rejects an additional stored identity-locus primitive unless ablation establishes its necessity. An address is not meaning, and a realization is not its constitution [M3, "Transformation-First Hypothesis" and "Semantic Locus Constraint"].

These distinctions shape the present proposal. They also constrain it. The prior Locus paper explicitly states that semantic binding does not by itself establish consciousness [M1, Section 14.5]. An experiential-binding thesis is consequently an extension requiring independent justification, not a result already contained in the earlier framework.

3.2 Geometry and structure are established research directions

Gardenfors's conceptual-space framework represents conceptual information through quality dimensions and geometrical relations [R2]. Representational similarity analysis compares patterns of dissimilarity across neural data, behavior,

and computational models [R3]. Neurophenomenal structuralism investigates constraints arising from relations between phenomenal and neural structures [R4]. These precedents already undermine any assertion that consciousness-related science considers only isolated, context-free neural objects.

The present proposal adds a particular emphasis on typed relational roles, contextual constitutions, transformation contracts, and provenance-sensitive comparison. Whether those additions are scientifically useful must be decided by ablation and predictive testing, not by the extensiveness of their vocabulary.

3.3 Olfactory and cooling research provide bounded footholds

Howard and colleagues reported correspondence between perceptual odor-quality similarity and distributed fMRI patterns in human posterior piriform cortex [R5]. McKemy and colleagues characterized a receptor activated by menthol and cool-to-cold thermal stimuli, providing a concrete peripheral mechanism relevant to a cooling component of mint-related sensation [R6]. De Araujo and colleagues investigated cognitive modulation of olfactory processing using different verbal descriptions of a test odor [R7]. Lee and colleagues developed a learned odor representation that supports prediction of odor qualities for previously uncharacterized molecules [R8].

These results concern particular mechanisms, perceptual judgments, or predictive representations. They do not establish that mint phenomenology is exhausted by one receptor, one region, one label, or one learned embedding. They do establish that the proposed program should build on existing empirical decompositions rather than portray the field as having no relevant structure to investigate.

3.4 Computation and contextual binding are not opposites

Integrated Information Theory 4.0 proposes an account connecting phenomenal properties to intrinsic cause-effect structure, not simply to an arbitrary matrix interpreted by an external observer [R9]. Hoffman and Prakash instead develop a consciousness-first mathematical framework with experiences among its primitives [R10]. The present proposal does not require either approach, and it does not establish their equivalence.

A contextual binding model can be computational, dynamical, mechanistic, or interpreted within another ontology. Computation can represent relations, history, boundaries, embodiment, and feedback. The claim that a context-free description is insufficient therefore does not establish that computational explanation is impossible.

Moreover, bare matrix indices are mathematically unlabeled in any theory. Their operational interpretation, state definition, and causal role matter. It would be unfair to demand explanatory power from an uninterpreted numerical array and treat its silence as a refutation of every functional theory.

3.5 Identifiability and causal abstraction

Unsupervised learning does not in general identify uniquely meaningful disentangled factors without additional assumptions; Locatello and colleagues make this limitation explicit in a formal and empirical analysis [R11]. Geiger and colleagues use causal abstraction and interchange interventions to test whether internal neural-network representations realize specified higher-level causal roles [R12].

These two results motivate complementary disciplines here: do not infer natural experiential axes from an arbitrary latent basis, and do not infer causal realization from successful decoding alone.

3.6 A division of explanatory responsibilities

The companion manuscripts do not all explain the same object. Their relevance is clearest when the research program separates a proposed account of experience from the instruments used to describe, test, and govern that account.

- **Source family:** Locus and Meaning as Binding [M1, M2, M4]
  Source-derived role: Relational semantic constitution, support, persistence, and recursion
  Application proposed here: Specify a contextual experiential-binding hypothesis
  Not supplied by the source: An established identity between semantic binding and felt quality
- **Source family:** GTK [M5]
  Source-derived role: Resolved binding identity, aliases, competing bindings, invariant floors
  Application proposed here: Keep category, source, episode, and record references distinct
  Not supplied by the source: A universal natural taxonomy of qualia
- **Source family:** ACO [M6]
  Source-derived role: Local descriptions of meaning-binding regimes under situational constraint
  Application proposed here: Candidate system-regime variables, separately operationalized
  Not supplied by the source: Calibrated mint-content coordinates or a phenomenal detection test
- **Source family:** Dynatology [M7]
  Source-derived role: Conceivable and admissible transitions, continuation, reachability
  Application proposed here: Bound which proposed experiential transformations have model realizations
  Not supplied by the source: Evidence that every conceivable quality profile is physically attainable

- **Source family:** Semantic Manifold Competence [M8]
  Source-derived role: Construction, maintenance, navigation, transformation, and governance capacities
  Application proposed here: Describe competence over experience-related representations
  Not supplied by the source: Proof that competence is experience
- **Source family:** SMT [M9]
  Source-derived role: Governance-relative auditing of declared preservation
  Application proposed here: Audit representation changes and whole-path contracts
  Not supplied by the source: A measured geometry or ontology of experience
- **Source family:** SMT worked example [M10]
  Source-derived role: Thick contextual relations and continuation-sensitive interpretation
  Application proposed here: Explain why an apparently simple descriptor opens a structured context
  Not supplied by the source: An independently validated physical or phenomenal model
- **Source family:** SBQ and its revision notes [M11, M12]
  Source-derived role: Binding integrity criteria and editorial alignment
  Application proposed here: Evaluate the quality of a research representation
  Not supplied by the source: Sensory intensity, pleasantness, or amount of consciousness
- **Source family:** CBT, CSC, and Meaning Crystals [M13-M15]
  Source-derived role: Anchor resolution, evidence/amendment, admission status, representational constraints
  Application proposed here: Make research commitments inspectable and revisable
  Not supplied by the source: A neural crystallization mechanism or proof that admission confers truth

This table is an integration proposal. It does not assert that the sources already implement a unified qualia workbench. In particular, conceptual vocabulary and software governance do not substitute for independent sensory or neural evidence.

3.7 ACO: complementary regime questions and a substantive boundary

ACO defines meaning-binding through constraint relevance and its persistence under transformation. It expressly distinguishes this usage from semantic, linguistic, or intentional meaning [M6, Section 2]. The expanded Locus manuscript

instead imposes a semantic threshold involving typed coherence, differential sensitivity, constraint, uptake, reidentifiability, jurisdiction, and consequence [M2, Section 4.4]. The two usages should therefore be represented as related but nonidentical predicates. A relation can modify continuation possibilities without thereby satisfying every condition of the Locus semantic threshold.

ACO's six local axes are Binding Density, Binding Integration, Binding Stability, Renegotiation Capacity, Meta-Binding, and Epistemic Autonomy. These characterize a system's binding regime, not the aromatic or cooling character of one episode [M6, Section 5]. The expanded Locus paper also distinguishes representation of a system's constraints from authority to revise them and from phenomenal consciousness [M2, Sections 10.4-10.6]. A detailed self-model is not, by that description alone, a source of either amendment authority or felt mint.

There is a genuine difference of target. ACO rejects qualia as a definitional boundary and characterizes them as private, unmeasurable claims, permitting them only as possible correlates of binding regimes [M6, Section 10]. The present paper does not silently replace that statement with a weaker claim. It adopts a different research commitment: public reports, comparisons, and interventions can constrain hypotheses about qualitative specificity, even though they do not provide direct, complete access to another experience. The proposed H-P hypothesis is an additional bridge, not a theorem of ACO.

The older S-to-C-to-I grammar in ACO also differs from the four-stage S/CM/IC/IR grammar in the expanded Locus account. In particular, ACO's Specification language already invokes admissibility, whereas the expanded Locus Specification includes constructible candidates that may be inadmissible [M6, Section 3; M2, Section 12]. This paper uses the latter grammar for research artifacts. It does not claim a lossless conversion of every historical use of the former.

3.8 Competence and continuation are different from qualitative content

Semantic Manifold Competence proposes intelligence as the capacity to construct, maintain, navigate, transform, and govern structured meaning while preserving coherence under novelty. Its scope statement explicitly leaves realizing mechanisms in place rather than claiming to replace them [M8, Section 2]. For the present program, remembering a distinction, comparing two reports, applying a category, or planning a transformation are competence tasks. Their success need not establish the phenomenal character of the event they concern.

Dynatology supplies a different object: a state space and relations distinguishing conceivable from admissible transitions [M7, Sections 3 and 8]. Its relevance is not that it has already found mint's physical dynamics. It permits a precise distinction between a describable target and an attainable continuation. Sections 9.6-9.10 use this distinction to formulate bounded realization queries.

The conversation worked example describes a thick origin-binding involving

context, history, identity, and future conversation. It also distinguishes a continuation being available from a specific assertion being warranted [M10, "Confidence Is Not Admissibility"]. This is an instructive template: the word *mint* can open a family of candidate distinctions without establishing that any one of them was experienced on a particular trial.

3.9 Continuity auditing is not the phenomenal territory

The main SMT manuscript defines its task narrowly: it checks whether declared semantic commitments survive authorized transformations. It expressly disclaims a theory of meaning and an unconditional metric geometry [M9, Abstract; Sections 3 and 21]. Its role here is consequently methodological. A chart conversion can be audited for preservation of recorded distinctions; that audit does not decide which distinctions a subject actually experiences.

There is a difference of scope between this definition and the worked example's statement that SMT describes structured reachability while GMB adds governance [M10, "Advanced Thesis"]. This revision retains the main manuscript's diagnostic definition when using the name SMT. The worked example supplies exposition, not an unnoticed replacement definition. Similarly, a failure of an authorized continuity claim does not by itself show that the underlying experience disappeared. Empirical absence and lack of audit standing have different types.

3.10 Binding quality and crystallization concern the research account

SBQ distinguishes binding presence from binding integrity and proposes five interacting dimensions: structural coherence, generativity, stability under perturbation, constraint alignment, and minimal distortion [M11, Sections 2-3]. These can motivate a research-model evaluation profile, but their operationalization for qualia studies is new work. The revision notes are editorial instructions, not independent confirmation of the SBQ theory; their terminology changes and unresolved overlay terminology are recorded as such [M12].

CBT contributes a loop from ambiguity through anchor resolution and realization to evidence and amendment, while warning against premature collapse and excessive fragmentation [M13, Sections 1 and 9]. CSC distinguishes a canonical artifact from one admitted to binding status and explicitly says admission does not guarantee correctness [M14, Sections 4-5]. Meaning Crystals examines representational constraints and "oracle surfaces," where an apparently explicit account still depends on undeclared interpretation [M15, Section 5].

Their proposed research use is to stabilize a testable account without erasing its uncertainty. A formal model can be frozen for evaluation while remaining a hypothesis. A raw report can be preserved when a coordinate schema cannot express it. Neither freezing nor admission turns a model into a conscious event. The source's crystallization language is not a thermodynamic mechanism [M14, Section 4.4]; the oscillatory language in Meaning as Binding and the semantic-energy analogy in SBQ likewise supply no measured neural bridge [M4, Section

8; M11, Section 4].

3.11 GMB and the universal governed semantic model

The **Governed Meaning Base (GMB)** is the current name in the author-confirmed historical lineage `CCF -> CMB -> GMB`. In this research program, GMB belongs to the architecture of the **research account**. It is not proposed as a neural substrate, a phenomenal field, or a container that stores qualia. A GMB could retain competing hypotheses, exact sources, observations, models, evidence, standing, counterevidence, and revision relations without making any retained claim true.

The universal governed semantic model can represent this paper through separately typed records:

- **Record:** experiential episode
  What it represents: the participant- and context-indexed target event
  What it must not be collapsed into: a report, category label, or database row
- **Record:** observation
  What it represents: report, comparison, behavior, physiological measure, or neural measure
  What it must not be collapsed into: transparent access to the episode
- **Record:** binding hypothesis
  What it represents: a proposed organization of roles, relations, context, and history
  What it must not be collapsed into: established phenomenal constitution
- **Record:** physical state
  What it represents: an independently characterized substrate state or trajectory
  What it must not be collapsed into: its measured projection or experiential descriptor
- **Record:** realization relation
  What it represents: a scoped mapping or intervention-sensitive bridge
  What it must not be collapsed into: identity merely because decoding succeeds
- **Record:** evidence
  What it represents: a result with method, scope, uncertainty, and alternatives
  What it must not be collapsed into: universal corroboration by the surrounding corpus
- **Record:** source
  What it represents: exact artifact and custody metadata

What it must not be collapsed into: logical-work identity, truth, or authority

- **Record:** standing
  What it represents: proposed, tested, supported, weakened, refused, superseded, or unresolved status
  What it must not be collapsed into: the underlying experience itself

This separation lets Hyperpedia or another pedagogical projection explain the program while retaining links back to exact evidence and unresolved alternatives. The projection may speak **about** a hypothesized experiential organization; it does not speak **from within** another subject's experience.

3.12 Corpus interfaces that govern the research account

Several later corpus works constrain how this proposal should be carried without becoming additional evidence for H-P:

- **Corpus interface:** *Semantic Legitimacy Before Context* [C1]
  Contribution here: require source, scope, exclusions, and interpretive authority before context is used inferentially
  Boundary: context volume is not legitimacy
- **Corpus interface:** *Reasoning by Way of Receipts* [C2]
  Contribution here: bind conclusions to premises, methods, evidence, failures, and claim ceilings
  Boundary: a receipt records reasoning; it does not manufacture truth or qualia
- **Corpus interface:** *The Three Hash Rule* [C3]
  Contribution here: distinguish source, revision candidate, and rendered or distributed artifact custody
  Boundary: byte identity is not scientific validity or authorship proof
- **Corpus interface:** *A Formal Theory of Work Custody* [C4]
  Contribution here: keep artifact, logical work, version family, authorship, and standing distinct
  Boundary: a successor candidate does not silently replace its source
- **Corpus interface:** *Admissibility Engineering* [C5]
  Contribution here: make study selection, evidence gates, privacy, and allowed effects explicit
  Boundary: an admissible study is not a confirmed hypothesis
- **Corpus interface:** *Semantic Debt* [C6]
  Contribution here: retain unresolved ambiguity, missing bridges, proxy risk, and interpretation burden as obligations
  Boundary: a polished vocabulary must not hide unpaid explanatory debt
- **Corpus interface:** *Legitimacy Without Self-Ratification* [C7]

Contribution here: require evaluation from commitments and evidence not minted by the model under test
Boundary: a workbench cannot confer its own scientific standing

These are typed governance interfaces around the research account. They do not establish that semantic binding constitutes experience and are not counted as independent corroboration of the paper's phenomenological claims.

4. The Contextual Experiential-Binding Hypothesis

Let an experiential episode be indexed by a subject or system a, an interval I, and an experimental jurisdiction J. A jurisdiction specifies the conditions under which a claim is intended to hold, including population, task, measurement procedure, and relevant environmental range.

Let $\mathcal{B}_e$ denote the hypothesized actually realized binding organization of episode e. Let $\mathcal{A}_e$ denote its architectonic description: the organized arrangement of components, relational roles, constraints, and transformation possibilities. Let Q_e denote the qualitative event, without assuming that it is directly measurable as a vector.

The central constitutive hypothesis is:

$$\boxed{\begin{array}{c}\text{The qualitative specificity of } Q_e \text{ is constituted by} \\ \mathcal{B}_e \text{ within its operative context and history.}\end{array}} \tag{1}$$

Equation (1) is a hypothesis in words, not an identity proved by notation. A mathematical symbol for binding does not provide a mechanism simply by being placed beside a symbol for experience.

4.1 Three nested commitments

H-S: structural adequacy. A context-indexed binding description captures reproducible distinctions in experience-related observations better than a flat vocabulary alone.

H-R: realization relevance. Some independently characterized substrate transformations systematically realize the changes predicted by a specified binding model, including under interventions and held-out contexts.

H-P: phenomenal constitution. The relevant realized binding organization constitutes qualitative specificity itself, rather than merely a correlated capacity to discriminate, categorize, remember, or report it.

H-S can hold while H-R fails. H-R can receive substantial support while H-P remains underdetermined. Even H-P, if established for qualitative specificity, would not automatically imply that any instance of semantic binding is sufficient for phenomenal existence.

The deliberately stronger universal thesis,

$$\text{semantic binding exists} \Longrightarrow \text{phenomenal experience exists}, \tag{2}$$

is not assumed. The present experiment does not test universal consciousness, panpsychism, or a global metaphysics of matter.

4.2 What the hypothesis excludes

For a specified implementation to be substantive, it must exclude possible outcomes. A model may, for example, predict that a relation-sensitive perturbation changes a particular quality profile even when component marginals remain approximately unchanged. It may predict that one context changes the effect of an intensity manipulation, while a matched irrelevant context does not. It may predict that a particular order of interventions matters.

If the same architecture can accommodate every result merely by adding an unmeasured contextual variable, it has ceased to function as an empirical explanation. Contextual boundaries, parameter counts, and revision rules must therefore be fixed before confirmatory testing.

5. Locus Is Not a Container: A Typed Ontology

5.1 Identity, support, and content

An identity address ι_e allows an episode, concept, or record to be referred to. It does not contain a quality. The support locus concerns the components and relations relevant to sustaining an event or a description of that event. The constitutional locus concerns invariants that define a proposed family of realizations. The realized event is neither its address nor its database entry.

A useful research record is:

$$D_e = \langle \iota_e, \widehat{\mathcal{A}}_e, c_e, h_e, J_e, \mathcal{R}_e, \mathcal{E}_e \rangle, \tag{3}$$

where $\widehat{\mathcal{A}}_e$ is an estimated architectonic description, c_e and h_e are declared contextual and historical variables, $\mathcal{R}_e$ specifies the proposed realization bridge, and $\mathcal{E}_e$ records evidence and uncertainty. Hats distinguish estimates from hypothesized underlying organization.

Equation (3) defines a descriptive artifact. It is not a recipe for manufacturing a conscious subject.

5.2 Typed binding hypergraphs

Represent a candidate architectonic by a typed attributed hypergraph:

$$G_e = (V_e, E_e, \tau_e, \rho_e, \alpha_e). \tag{4}$$

Here V_e contains candidate components; E_e contains relations that may involve several components; τ_e assigns types; ρ_e assigns participant roles within each relation; and α_e contains attributes such as values, units, time intervals, and uncertainty.

A binding involving sensory input, expected source, and perceived location is not equivalent to an unordered bag containing those three items. Role assignments are part of the hypothesis. A component serving as a current sensory input is not interchangeable with the same component represented as a memory or an instruction.

Each modeled binding also carries an epistemic descriptor:

$$\begin{aligned} \widehat{b} = \langle &\text{relata}, \text{roles}, \text{context}, \\ &\text{interval}, \text{constraint}, \text{provenance},. \\ &\text{standing} \rangle \end{aligned} \tag{5}$$

Provenance and standing belong to the research account, not necessarily to the organism's mechanism. It would be a category mistake to claim that a neuron needs an administrative authorization record before a sensation can occur.

5.3 Bounded support rather than universal context

A proposed support set should be no larger than its explanatory role warrants. Relevant history may be encoded in present adaptation or memory states; it need not be a complete chronological archive. A cultural label may influence a task through expectation without being a necessary constituent of every sensory event. An external mint leaf may be a stimulus source without being part of every current internal realization after the leaf is removed.

For a declared model class and intervention family $\mathcal{I}_J$, define an operational support as a set of variables sufficient for a specified predictive or interventional task. Removing a variable that changes neither task performance nor the claimed intervention structure supplies evidence against its necessity *within that task*. It does not prove universal dispensability.

This formulation avoids converting distributed support into the assertion that everything in the universe participates equally in every quale.

5.4 Recursive and reflexive binding

A subject may experience a cooling sensation and subsequently recognize it as mint, associate it with toothpaste, or evaluate the recognition itself. These can be modeled as successive or interacting bindings:

$$\mathcal{B}_0 \longrightarrow \mathcal{B}_1(\mathcal{B}_0, \text{category}) \longrightarrow \mathcal{B}_2(\mathcal{B}_1, \text{self-assessment}). \tag{6}$$

The notation specifies higher-order organization, not a compulsory ladder that every experience must ascend. Elementary experience need not include a reflective concept of itself. Experimental reporting may introduce higher-order bindings that were not present in the original event; this is a measurement issue, not an inconvenience to ignore.

5.5 Identity has several noninterchangeable targets

GTK treats the resolved semantic binding as the lexical unit, accompanied by aliases and competing bindings; its invariant floor attaches to that binding, not to the raw word [M5, Sections 1-3]. Applying that structure to the research registry requires a further distinction between a rich lexical entry and the opaque address used to retrieve it. The entry can contain semantic commitments; an address alone cannot supply them.

A proposed identity discipline distinguishes at least five object kinds: a category such as mint-related character; a preparation or source; a delivered-stimulus event; an experiential episode; and a measurement or documentary record. A sixth kind, the binding hypothesis, describes relations among these without being identical to any of them. Thus two trials can share a preparation and category while differing in delivered stimulus, episode identity, or reported character.

The principal relations are also typed:

$$\begin{aligned} \text{SameCategory}_J(e_1, e_2) &\not\Rightarrow \text{SameEpisode}(e_1, e_2), \\ \text{SameSource}(e_1, e_2) &\not\Rightarrow \text{SameProfile}_J(e_1, e_2), \\ \text{MatchedObservations}_J(e_1, e_2) &\not\Rightarrow \text{PhenomenallyIdentical}(e_1, e_2). \end{aligned}$$

These are restrictions on inference, not claims that the pairs can never coincide. An aftertaste continuing through time can retain one episode identifier while its measured profile changes. A repeated mint exposure can generate a different episode with a similar profile. The study must declare its temporal individuation rule rather than letting database row boundaries define experience.

A content digest identifies particular serialized bytes or a declared canonical form. A stable reference identifies the record lineage or entity to which revisions refer. A new digest need not mean a new subject or a new experience. Conversely, reusing a reference does not prove that the underlying binding constitution was preserved. This separation is compatible with GTK's content-addressed

entry revisions [M5, Section 4], but must be implemented explicitly rather than treating every kind of identity as one hash.

5.6 Three loci in the mint example

The source distinction among constitutional, standing, and realization loci [M1, Section 6; M2, Section 5.4] supports a useful experiential application. A proposed constitutional locus specifies the invariants of a bounded family of episodes. A proposed standing locus comprises relevant dispositions or preserved conditions for later reactivation. A proposed realization locus identifies the actual support of a particular ongoing episode. These are hypotheses about different explanatory roles, not three necessarily disjoint physical regions.

A memory of mint, the ability to recognize mint, and a present mint-related sensation can consequently share structures without being treated as one event. Nor does semantic recurrence prove that the same phenomenal quality has recurred. The research task is to identify which relations remain stable, which change, and which distinctions the available observations can support.

Provenance, an investigator's approval, and the validity of a measurement belong to the account of these loci. They should not be inserted into the proposed biological mechanism unless a separately specified causal pathway makes them relevant, for example through an instruction that changes a participant's expectation. The mere existence of an approval record is not such a pathway.

6. Mint as a Family of Contextual Realizations

6.1 A multidimensional target without invented coordinates

The everyday phrase *taste of mint* is retained for continuity with the motivating question. Operationally, the research target should distinguish possible gustatory, olfactory, cooling or irritation-related, oral-somatosensory, and temporal contributions. The menthol-sensitive cold-receptor finding supplies one particular physiological foothold, not a reduction of the entire experience to cooling [R6].

Candidate descriptive dimensions include perceived overall intensity, cooling, aromatic character, sweetness, bitterness, perceived freshness, oral localization, texture, temporal persistence, familiarity, and pleasantness. Attention, expectation, recent exposure, and bodily state may be modeled as context, as components of wider experiential organization, or both under explicitly different analyses. They should not silently change roles between training and testing.

These candidate dimensions are not established orthogonal primitives. Some may be redundant, composite, culturally variable, inaccessible to a particular measurement, or irrelevant in a given jurisdiction. A sparse descriptor is preferable to an elaborate inventory that cannot be independently assessed.

6.2 Contrasts and their interpretation

- **Contrast:** Weak versus strong mint
  Candidate change: Intensity trajectory, potentially with quality and irritation changes
  What must not be assumed: A concentration increase affects only one experiential dimension
- **Contrast:** Peppermint versus spearmint
  Candidate change: Discriminable aromatic and cooling profiles under a specified preparation
  What must not be assumed: A universal numerical distance or fixed sweetness ranking
- **Contrast:** Fresh versus dried mint
  Candidate change: A combination of preparation, release, texture, and contextual differences
  What must not be assumed: Freshness is a single independent physical variable
- **Contrast:** Fresh versus decay-associated mint
  Candidate change: Preservation of some identity cues with additional aversive or unfamiliar qualities
  What must not be assumed: Rotting material is safe to taste or a one-axis stimulus manipulation
- **Contrast:** Onset versus aftertaste
  Candidate change: Time-dependent change, persistence, and adaptation
  What must not be assumed: A static average adequately describes the event
- **Contrast:** Same preparation under different expectations
  Candidate change: A possible change in categorization, report, or experience
  What must not be assumed: Any report change proves a phenomenal change

The rows are experimental contrasts, not claims that their mechanisms or outcomes are already known. Botanical identity, preparation, chemical composition, delivered stimulus, category label, and phenomenal character should be recorded separately.

For decay-associated cases, the study should use only institutionally approved safe representations or substitutes. The conceptual contrast does not authorize tasting spoiled material or exposing participants to uncontrolled biological hazards.

6.3 Mint as a region, not necessarily an atom

Let $\mathcal{Q}_{a,J}$ be a provisional space of within-subject experiential distinctions under jurisdiction J. A lexical mint category can be modeled as a region:

$$\mathcal{M}_{a,J} \subseteq \mathcal{Q}_{a,J}. \tag{7}$$

The boundary may be uncertain, context-sensitive, or graded. It is not assumed to be convex, connected, or identical across people. An unfamiliar sensation may be representable without a stable lexical category. Conversely, two discriminably different events may receive the same label.

A particular episode is represented, where supported, by a trajectory:

$$q_e : I \to \mathcal{Q}_{a,J}. \tag{8}$$

This is a coordinate-level representation of the event, not the event itself. Point estimates should be replaced by uncertainty distributions when the data warrant no unique placement.

7. Orthogonal Dimensions: Four Different Claims

The expression *orthogonal dimensions* can be valuable only if its meaning is controlled.

7.1 Typed dimensional distinction

Two properties can be separately named and assigned distinct measurement procedures. Aromatic identity and cooling intensity are conceptually distinguishable. This is a type distinction, not proof that either can vary independently of the other.

7.2 Statistical independence and decorrelation

For measured descriptors q_i and q_j, zero covariance is not generally equivalent to independence:

$$\mathrm{Cov}(q_i, q_j) = 0 \implies p(q_i, q_j) = p(q_i)p(q_j). \tag{9}$$

A rotated or whitened coordinate system can have uncorrelated components without revealing independent sensory mechanisms. The learned basis also depends on sampling, task, and model assumptions [R11].

7.3 Geometric orthogonality

Where a differentiable local chart and a justified metric g exist, tangent directions u and v are orthogonal when

$$u^\top g(q) v = 0. \tag{10}$$

Neither the metric nor its interpretation is supplied by writing a feature list. It might be derived from a specified discrimination model, with uncertainty and a local domain of validity. Euclidean geometry is an optional model, not a default fact about experience.

7.4 Causal selectivity

Let u_k denote a controlled intervention parameter and $\widehat{q}_i$ an estimated coordinate. A local intervention-response matrix is

$$J_{ik} = \frac{\partial \mathbb{E}[\widehat{q}_i \mid do(u)]}{\partial u_k}. \tag{11}$$

A manipulation aimed at dimension k is selectively effective only relative to declared on-target and off-target criteria. A possible diagnostic is

$$L_k = \frac{\sum_{i \notin T_k} w_i |\Delta \widehat{q}_i|}{\epsilon + \sum_{i \in T_k} w_i |\Delta \widehat{q}_i|}, \tag{12}$$

where T_k is the intended target set and scales, weights, and ϵ are fixed in advance. A low value does not prove perfect independence; a high value can reveal coupled dimensions, a poor intervention, or an incomplete model.

Selectivity also requires a nontrivial on-target effect. A manipulation that changes nothing should not receive a high success score simply because it has no off-target effects.

7.5 Dependency is a finding, not a failure of the whole program

Increasing a delivered mint stimulus might alter intensity, cooling, quality, or pleasantness together. Such coupling does not automatically refute a binding framework. It does refute a particular claim of selective control or separability when that claim was preregistered.

Conversely, the framework must not become immune to falsification by treating every coupling as a victory. The empirical commitments concern a specific graph, dimension set, and transformation family. Revisions must be recorded as revisions and evaluated on new data.

7.6 Content, regime, competence, and evidence are different coordinate families

The source integration introduces four descriptive families:

$$\mathcal{D}_e = \langle q_e, r_e, k_e, \varepsilon_e \rangle.$$

Here q_e describes the episode's candidate qualitative profile; r_e describes the system's candidate binding regime; k_e describes competence on declared operations; and ε_e describes the evidence available to the investigator. This tuple is bookkeeping, not an assertion of a product geometry, causal independence, or full observability.

For r_e, the source vocabulary is ACO's six axes. For k_e, it is the construction, maintenance, navigation, transformation, and governance capacities of Semantic Manifold Competence [M6, Section 5; M8, Section 2.1]. Values must remain unassigned until independently operationalized. A number of self-referential sentences is not automatically a measurement of Meta-Binding; a persistent category label is not automatically a measurement of Binding Stability.

The four families can interact. A modeled capacity to retain distinctions might affect a delayed report, and an expectation might affect a content judgment. These are candidate pathways to test, not reasons to collapse the variables into one scale. A richer description of a system's regime does not entail stronger mint, and a more complete evidence record does not entail a more intense or more conscious episode.

A useful comparative design therefore asks whether independently measured regime or competence variables explain additional variance or transfer beyond sensory conditions and history-aware baselines. It separately tests whether any apparent gain is confined to reporting. No six-axis threshold is proposed as necessary or sufficient for mint experience.

8. Experiential Atlases, Not an Assumed Universal Manifold

8.1 Local charts

A chart is a bounded representational procedure:

$$\phi_{a,J} : \mathcal{E}_{a,J} \to \mathcal{Z}_{a,J}, \tag{13}$$

where $\mathcal{E}_{a,J}$ is a class of episodes and $\mathcal{Z}_{a,J}$ can contain continuous coordinates, ordinal categories, discrete relations, and missing values. A conventional smooth manifold is a special case, requiring additional mathematical and empirical justification.

For example, intensity ratings might be modeled ordinally while a stimulus-source relation is categorical and timing is continuous. It would be inappropriate

to force all three into an unqualified Euclidean distance just because software accepts a numeric array.

8.2 Cross-context and cross-person maps

When two charts overlap, a transport map may be estimated:

$$T_{J\to K} : \mathcal{Z}_{a,J} \dashrightarrow \mathcal{Z}_{a,K}. \tag{14}$$

The dashed arrow denotes a partial map: some states may not transport, and uncertainty can make the image a distribution rather than a point. A cross-person map adds an independently calibrated subject index. It should not be inferred from identical verbal labels alone.

Transport is evaluated by preservation of declared relations, rankings, or intervention responses on held-out episodes. Failure to establish a map is an informative result. It may reflect measurement incompatibility, inadequate data, individual differences, or a genuine limit of the shared model.

8.3 Compatibility without forced gluing

The Locus framework's atlas and presheaf vocabulary suggests comparing local descriptions on their overlaps [M1, Section 7]. A disciplined use here is simply to require that two transported descriptions agree on shared measured quantities within declared tolerances before treating them as one coherent account.

A full sheaf formulation would additionally require a specified context structure, restriction maps, and a justified gluing condition. Those are not established merely by drawing overlapping circles. Nor does failure to glue constitute evidence of quantum contextuality, nonlocal consciousness, or a nonphysical substrate.

8.4 Historical dependence and path order

Let T_1 and T_2 be context-changing or stimulus-changing operations. A dynamic binding model may predict

$$T_2 \circ T_1(q) \neq T_1 \circ T_2(q). \tag{15}$$

An order effect can motivate a history-sensitive model, but does not uniquely support the present framework. Adaptation, memory, response strategies, and other dynamical explanations must remain active competitors. The binding account earns support only when its specified organization predicts the order effect more accurately or parsimoniously than those alternatives.

A trajectory need not be Markovian in the chosen descriptors. If omitted history matters, one can augment the state or use an explicit history-dependent model.

State augmentation is a modeling operation, not a proof that the resulting state is phenomenally complete.

9. The Realization Problem

9.1 Physical state, measured state, and experiential descriptor

Let z_t denote the relevant physical state at a declared resolution. Let x_t denote a measurement of it:

$$x_t = M(z_t, \eta_t), \tag{16}$$

where M is an instrument model and η_t includes measurement noise and nuisance factors. Let y_t denote experience-related observations such as a rating, a comparison judgment, a choice, or a timed report.

The proposed realization relation is

$$\mathcal{R}_J \subseteq \mathcal{Z}_{\text{physical},J} \times \mathcal{B}_J \times \mathcal{E}_J. \tag{17}$$

It relates physical support, binding organization, and an experiential episode. It is written as a relation rather than a function because many-to-one realization, incomplete specification, and uncertainty are all possible. A function becomes appropriate only after the necessary sufficiency assumptions are stated.

The operationally estimable object is more modest:

$$\widehat{p}(y_{t:t+\Delta} \mid x_{\leq t}, c_{\leq t}, h_{\leq t}, u_{\leq t}). \tag{18}$$

This predicts observations under specified conditions. It is not automatically an estimator of an independently accessible phenomenal state.

9.2 What it means to plot electronics or neurons

The initiating proposal asks for structures that permit plotting electrical or neural changes corresponding to qualitative realization. The appropriate physical variables depend on scale: a computational model's state variables, circuit voltages, ion-channel dynamics, population activity, or an instrument's indirect measurement. The proposal does not require tracking every electron, and changes in neuronal activity should not be described as whole neurons moving through a semantic space.

Instrument variables are not interchangeable with microscopic mechanisms. A bridge from fMRI patterns to reported odor similarity, for example, is a bridge at the measurement level unless further mechanistic evidence is supplied. The spatial and temporal transformations involved in neural measurement are explicitly relevant to representational analysis [R3].

A useful model may operate at a coarse level. Its usefulness does not license claims about finer-scale physical causes it did not measure.

9.3 No requirement of global distance preservation

It is tempting to demand

$$d_{\text{physical}}(z_i, z_j) \approx d_{\text{experience}}(q_i, q_j). \tag{19}$$

But even a correct realization relation need not preserve raw distances. Redundant microstates can realize the same higher-level condition. Measurement can distort distances. The relation can be nonlinear, history-dependent, and subject-specific.

The stronger scientific target is therefore not global geometric resemblance but *specified transformation correspondence*. A physical manipulation should have the high-level consequences predicted by the bridge, with known bounds and exceptions.

9.4 Intervention-commutation

Let M_μ be a low-level causal model and M_B a higher-level binding model. Let ρ map low-level states into higher-level variables, and let ω map an allowed low-level intervention i into its hypothesized high-level counterpart. A realization test compares

$$\rho_* P_{M_\mu}^{do(i)} \quad \text{with} \quad P_{M_B}^{do(\omega(i))}, \tag{20}$$

where ρ_* denotes the induced distribution after mapping. The disagreement can be quantified as

$$\mathcal{E}_{\text{comm}}(i) = D\left(\rho_* P_{M_\mu}^{do(i)}, P_{M_B}^{do(\omega(i))}\right), \tag{21}$$

using a divergence or distance appropriate to the variables and sampling design. Causal-abstraction methods provide a precedent for testing such higher- and lower-level alignments [R12].

A small commutation error supports the tested correspondence. It does not by itself establish that the high-level variables exhaust experience. Interventions, mappings, and targets must be specified independently of the held-out outcomes; otherwise an exact fit can be engineered by construction.

9.5 No guaranteed reverse compiler

A forward prediction model does not imply that any desired quality profile can be induced. A proposed inverse problem is

$$u^* = \arg \min_{u \in \mathcal{U}_{\text{allowed}}} \mathbb{E}\left[d(\widehat{q}_{t+\Delta}, q^*) \mid do(u)\right] + \lambda \operatorname{Cost}(u). \tag{22}$$

This formulation requires a causal model, a feasible target, and an independently authorized intervention set. The solution may be nonunique or nonexistent. A structurally incompatible target must be refused rather than approximated while reporting success.

The word *compiler* is thus a useful analogy for a validated translation between levels, not evidence that experiences can already be programmed.

9.6 Dynatological reachability under a declared physical model

Following Dynatology's transition-system core, let

$$\mathcal{D}_{\mu,J} = (Z, \Omega_\mu, A_\mu)$$

be a bounded model of physical or simulated states under jurisdiction J. Here Ω_μ records model-conceivable transitions and $A_\mu \subseteq \Omega_\mu$ the transitions admitted by the specified dynamical model. Where actions matter, write A_μ^u for the transition relation under action u. None of these symbols asserts that all relevant physical states or actions have been discovered [M7, Sections 3, 8, and 25].

Physical feasibility, evidentiary availability, and execution permission remain separate. A researcher may lack evidence for a feasible transition. A well-supported feasible transition may be outside the authorized protocol. A permitted model query may return no path. The graph used for a particular execution request can intersect these constraints, but the record must retain the result of each check rather than call every excluded edge physically impossible.

For finite nonnegative edge costs, define a directed path cost by minimizing their sum over admitted paths. No path gives infinite cost within the declared model and horizon. This is not asserted to be a symmetric phenomenal distance. A cheaply reachable descriptor can be very dissimilar; a similar descriptor can be unreachable by the available controls. A permitted zero-step identity path also does not establish indefinite viability.

9.7 A projected path can be a false promise

Let $\rho : Z \to \widehat{Q}$ be a fixed map into a space of modeled experiential descriptors. An existentially projected transition is

$$(q, u, q') \in A_Q \quad \Longleftrightarrow \quad \exists z,$$
$$z' \in Z : \rho(z) = q,\ \rho(z') = q',\ (z, z') \in A_\mu^{u\cdot}$$

This is a useful coarse representation, but it can join witnesses that do not join in the underlying dynamics. Consider four states with exactly these relevant transitions:

$$a \xrightarrow{u} b, \qquad c \xrightarrow{v} d,$$

and projections

$$\rho(a) = q_0, \quad \rho(b) = \rho(c) = q_1, \quad \rho(d) = q_2.$$

The projected graph contains $q_0 \xrightarrow{u} q_1 \xrightarrow{v} q_2$. Nevertheless, no corresponding two-action path starts at a: the first step ends at b, whereas the second edge has witness c. Equality of a descriptor has been mistaken for interchangeability of underlying states.

This counterexample is a new application of the supplied transition and path-governance ideas, not an empirical finding about mint. In a future experiment, the concealed difference might be an independently measured adaptation state or other history variable. The example itself does not establish which variable matters.

An extra transition $b \to c$ would not retroactively make the original two-step sequence valid. It would be an additional action or dynamical interval requiring its own timing, costs, and evidence. Likewise, an experimental reset between separately validated transitions must appear in the complete path. It cannot be silently inserted by the analysis.

9.8 Complete-path realization and its quantifiers

For a descriptor/action sequence

$$\pi_Q = (q_0, u_0, q_1, \ldots, u_{n-1}, q_n)$$

and a declared initial-state set Z_0, define compatible endpoint sets recursively:

$$L_0 = Z_0 \cap \rho^{-1}(q_0),$$

$$L_{j+1} = \left\{ z' \in \rho^{-1}(q_{j+1}) : \exists z \in L_j,\ (z, z') \in A_\mu^{u_j} \right\}.$$

A nonempty L_n witnesses at least one complete model path consistent with the specified sequence and initial set. The qualifier *at least one* matters. The actual starting state may not be the successful member of Z_0. Nondeterministic dynamics may also admit failing outcomes alongside the witness. A robust-control or probabilistic-success claim requires a different quantified test and, for probabilities, a calibrated stochastic model.

The recurrence addresses fixed finite paths with endpoint-state sufficiency. A cumulative resource limit or history-dependent obligation requires an augmented state or retained path monitor. If two individually affordable transitions cost two units each but the whole-path budget is three, the sequence fails the budget even when its physical endpoint sequence exists. SMT's independent path-level contract is directly relevant to documenting this distinction [M9, Section 11].

A complete bounded model with no witness warrants *no lift in this model*. An incomplete observed graph with no witness warrants an unresolved realization query, not physical impossibility. A model witness is also not an observed physical execution. These are separate evidence levels.

9.9 Continuation-sensitive refinement

A descriptor map intended for sequential prediction must retain enough state to preserve the continuations relevant to that prediction. One strong sufficient condition is the local lifting condition:

$$\begin{aligned} &\forall z, u, q' : \ (\rho(z), u, q') \in A_Q \\ &\implies \exists z' : \ (z, z') \in A_\mu^u \\ &\wedge \ \rho(z') = q' \end{aligned} \qquad .$$

Under that condition, every finite projected path can be lifted from every compatible initial state, by successive selection of witnesses. The condition is stronger than existential projection and need not hold in realistic coarse charts. It is a sufficient modeling contract, not a universal requirement for useful descriptive coordinates.

When it fails, the research program has several options: restrict the chart's claim to description; retain a distribution over hidden states; augment the state with independently justified history; or refine a merged descriptor into continuation-distinct classes. These options must compete on frozen sequential holdouts. Adding a hidden variable after every failure would make the account nonpredictive.

A crucial consequence is that *phenomenologically matched according to current measurements* need not mean *causally interchangeable for the next intervention*. A content chart can be adequate for similarity judgments while inadequate for intervention planning. Rejecting its use for planning does not establish that its descriptive distinctions are false.

9.10 Viability is an optional dynamical property, not a phenomenal verdict

Dynatology defines viability through the existence of an infinite admitted continuation [M7, Section 12]. For a finite model, this is equivalent to reaching a directed cycle, including a self-loop. The source's fixed-point deletion rule supports this definition; its separate closed-component algorithm wording requires the qualification recorded in Appendix E.

This criterion is existential. A state with a self-loop and an exit to a terminal state is viable in that sense because continuation can remain on the loop. It is not necessarily safe under every controller or almost surely persistent under stochastic selection. Nor is a terminal state conscious or unconscious because of its graph position. The optional computation is useful only for explicitly declared questions about sustained model continuation.

The finite fixture in the companion brief checks this distinction and the path counterexample. It does not analyze an actual neural viability kernel or provide a biological survival prediction.

10. Measurement: Experience Is Not Its Report

10.1 A measurement model

Suppose q_e is a latent coordinate representation of an episode. A report model may be written

$$y_e \sim p_\theta(y \mid q_e, \ell_e, d_e, k_e), \tag{23}$$

where ℓ_e is the response vocabulary, d_e the decision setting, and k_e the reporting task. A context manipulation can affect these variables without necessarily changing the underlying experience. Conversely, experience can change without a particular response instrument detecting it.

This is why label effects are relevant but not decisive. Research on cognitive modulation of olfactory processing motivates controlling verbal context [R7]; it does not authorize treating every verbal difference as a direct readout of phenomenology.

10.2 Triangulation without fictitious access

The proposed battery combines several imperfect measures: repeated intensity and quality ratings, pairwise or triadic similarity judgments, discrimination choices, temporal descriptions, confidence assessments, and selected physiological or neural observations. Their convergence can strengthen an interpretation. Their disagreement must be retained.

No channel is automatically a phenomenal ground truth. A classifier output is not an independent witness when it was trained on the same labels used to

define the target. Multiple questions answered in one response session are not independent observers merely because they occupy different columns in a file.

Within-subject calibration is the initial priority. Cross-subject alignment is an additional hypothesis with its own validation burden. A shared word, shared stimulus, or shared coordinate value is insufficient evidence of identical experiences across people.

10.3 Representational residuals and oracle surfaces

Meaning Crystals identifies schema insufficiency, unrepresentable concepts, and undeclared interpretive assumptions as possible oracle surfaces [M15, Section 5]. Applied to measurement, the practical requirement is to preserve a participant's original description, its elicitation procedure, and the normalization that produced a coordinate record. A phrase that does not fit the current atlas is a representational residual, not automatically noise or zero intensity.

The study should distinguish a participant declining to report, being unable to use the requested vocabulary, having an inapplicable dimension, and supplying a description that the schema cannot encode. These conditions have different evidentiary implications. Original descriptions are not themselves infallible phenomenal ground truth; retaining them makes interpretation inspectable rather than resolving the problem by decree.

Alternative schemas can be compared for held-out reconstruction of declared distinctions, discriminatory usefulness, and uncertainty calibration. A schema that silently deletes residuals may appear more stable precisely because it cannot represent its own failures.

10.4 Model identification and sensitivity

The analysis should compare alternative measurement models, including models in which contextual effects act predominantly through response criteria. A phenomenal interpretation gains credibility only to the extent that it survives plausible alternatives and predicts evidence not used to construct it.

A useful sensitivity analysis changes the assumed reliability of reports, permits subject-specific scale usage, and tests how much unmeasured confounding would be required to explain the apparent bridge. Where several latent models fit equally well, the appropriate output is a set of admissible explanations, not an artificially precise latent coordinate.

11. Limited Formal Results

The following propositions are mathematical consequences of their assumptions. They are not empirical results about mint and are not asserted to be novel theorems.

11.1 Proposition 1: Address-renaming invariance

Let D be a well-formed descriptive artifact. Suppose an injective renaming σ changes identity addresses and all corresponding references while preserving typed roles, values, and relationships. If the prediction and validation procedures depend only on that preserved structure, then

$$F(\sigma D) = F(D). \tag{24}$$

Proof. By assumption, every argument inspected by F is preserved, and addresses have no independent semantic payload. Thus the renaming does not change any relevant input. The outputs are equal. If the artifact is serialized differently, its ordinary file hash can nevertheless change. □

This proposition operationalizes the distinction between identity addressing and content. A model that uses accidental record identifiers to improve prediction violates the intended invariance and may be exploiting leakage.

11.2 Proposition 2: Observational coordinates are not uniquely identified

Suppose a latent model has state q, transition law K, and observation law $p(y \mid q)$. For any invertible transformation g, define $q' = g(q)$, transform the transition law by g, and set

$$p'(y \mid q') = p(y \mid g^{-1}(q')). \tag{25}$$

The transformed model induces the same observational distribution.

Proof. Substitute $q = g^{-1}(q')$ in the latent-state construction and transport the initial and transition distributions accordingly. The observed outputs retain their original law. □

Therefore, an observationally successful embedding does not uniquely identify intrinsic mint coordinates. Calibration, constraints, and interventions can restrict this freedom but do not remove every interpretive question. This elementary argument is consistent with broader identifiability concerns in representation learning [R11].

11.3 Proposition 3: Local controllability is conditional

Assume a differentiable, locally valid map $f : u \mapsto \mathbb{E}[\widehat{q} \mid do(u)]$ with Jacobian J at u_0. A small desired change δq is achievable to first order by an unconstrained local perturbation only if

$$\delta q \in \mathrm{im}(J). \tag{26}$$

Proof. First-order changes have the form $J\delta u$. Their set is the image of J. □

Even membership in this image is not sufficient for real-world feasibility: safety bounds, parameter limits, nonlinear effects, and uncertainty may preclude the intervention. The proposition supports refusal of unsupported inverse claims, not a promise of controllable qualia.

11.4 Proposition 4: Matched marginals need not determine binding-sensitive outputs

Let two variables a and b occupy different roles and define

$$F(a, b) = (a, b, ab). \tag{27}$$

Then $F(a, b) \neq F(b, a)$ when $a \neq b$, although the unordered multiset of component values is unchanged.

This elementary counterexample shows why an unordered component inventory need not determine a role-sensitive output. It does not show that actual mint experience has this form. The empirical question is whether a similarly specific relational distinction improves prediction under controlled conditions.

11.5 Operational equivalence is not phenomenal identity

For an exactly specified intervention family, two implementations may be called operationally equivalent when they yield the same distributions over all declared observables under corresponding interventions. That is a genuine equivalence relation under suitable fixed definitions.

Approximate indistinguishability within a tolerance is different. It need not be transitive: A can be close to B and B close to C while A is not close to C. Therefore, approximate pairwise similarity must not automatically trigger identity merging.

Neither exact operational equivalence on a restricted test family nor approximate similarity establishes phenomenal identity. The claim remains bounded by the observation model and the interventions actually considered.

11.6 Proposition 5: Finite-path lifting criterion

Fix Z, A^u_μ, ρ, Z_0, and a finite descriptor/action sequence as in Section 9.8. Assume all constraints relevant to continuation are represented in the state. There exists a low-level path beginning in Z_0 that projects to the entire sequence if and only if $L_n \neq \varnothing$.

Proof. At step zero, L_0 is exactly the compatible initial states. Inductively, $z' \in L_{j+1}$ precisely when there is a state in L_j with an allowed u_j transition to z' and the required next descriptor. By the induction hypothesis, that predecessor has a witnessing prefix. Extending the prefix supplies a witness for z'. Conversely,

the endpoint of every witnessing prefix is inserted by the recurrence. The claim follows at step n. □

The proof establishes a decision rule for a stipulated finite model. It does not establish the completeness of a measured graph. With cumulative nonnegative cost as the only additional path variable, retaining the least-cost witness per endpoint suffices for a fixed budget. More general history-dependent obligations require an appropriate monitor state or multiple nondominated histories.

11.7 Proposition 6: Edgewise realization is insufficient

Existence of a witness for each projected edge does not imply a compatible witness for the concatenated path.

Proof. In Section 9.7, the first edge is witnessed by (a, b) and the second by (c, d). The only permitted first-step endpoint from a is b, which has no v successor. Thus $L_1 = \{b\}$ and $L_2 = \varnothing$, although both projected edges exist. □

This is a standard kind of coarse-state ambiguity expressed here as a realization contract. No priority claim is made for the elementary graph result. Its role is to expose an inference that an experiential "compiler" must not make.

11.8 Corollary: A strong local lifting contract suffices

Under the condition of Section 9.9, every finite projected path lifts from each compatible initial state.

Proof. Start from that state. At each step the condition provides a successor in the required next fiber. Finite induction constructs the full path. □

This is not a robustness theorem: other successors may violate the requested sequence. Existential reachability, universal nondeterministic safety, and probability of successful execution require distinct contracts.

12. Empirical Predictions and Competing Explanations

12.1 Prediction family P1: Context-sensitive generalization

A preregistered binding model predicts how specified contextual variables modify responses to the same delivered stimulus. It must outperform not only a context-free stimulus model but also a strong contextual baseline with comparable information and modeling capacity.

Support: improved held-out predictive performance and calibration, particularly in contexts or combinations omitted from training. **Challenge:** no practically relevant improvement, or gains eliminated by a simpler contextual model. A label effect by itself supports neither a uniquely binding-based account nor H-P.

12.2 Prediction family P2: Compositional transformation transfer

A model fitted to selected aroma, intensity, and context combinations predicts new combinations using shared rules rather than a separate entry for every condition.

Support: successful held-out composition with interpretable limits. **Challenge:** performance collapses outside familiar combinations, or a flat matched-capacity model transfers just as well. The claim is about explanatory economy and transfer, not an assertion that ordinary neural networks cannot learn interactions.

12.3 Prediction family P3: Binding-sensitive relational contrasts

Construct conditions that preserve relevant component distributions as closely as feasible while changing their source, temporal, spatial, or expectation-related binding. The model predicts a specific change in reported organization or discriminability.

Support: predicted relational effects after controls for stimulus delivery, attention, timing, and reporting. **Challenge:** the effect disappears under controls or is fully accounted for by unmodeled low-level differences. Perfect physical matching is difficult; residual differences must not be hidden beneath the phrase *same components*.

12.4 Prediction family P4: Intervention correspondence

An independently selected intervention has a prespecified effect on a binding variable and its downstream measured consequences. The bridge is tested on interventions not used to fit its correspondence.

Support: low held-out commutation error with acceptable on-target effect and bounded off-target changes. **Challenge:** predictive decoding remains high while intervention correspondence fails. This dissociation would show that a correlational representation was mistaken for a realization map.

12.5 Prediction family P5: Temporal and historical organization

A dynamical model predicts onset, persistence, adaptation, and order effects better than static summaries and competitive history-aware baselines.

Support: transfer to new temporal schedules without adding a parameter for each schedule. **Challenge:** gains disappear after accounting for simple decay or adaptation, or the supposedly necessary history terms have no measurable contribution.

12.6 Prediction family P6: Restricted cross-realization correspondence

A transformation defined at the high level can have distinct lower-level implementations while preserving the tested causal profile. A computational fixture,

a learned neural model, and human measurements can be studied as separate tracks.

Support: bounded correspondence under independently defined tasks and interventions. **Challenge:** a shared semantic label is the only commonality, or alignment works only after outcome-dependent remapping. Even successful correspondence would not establish that all implementations share phenomenal character.

12.7 Prediction family P7: Sequential realization and hidden-state discrimination

A preregistered model predicts complete sequences rather than only isolated changes. It must distinguish conditions with matched current descriptors but independently characterized differences in future response. Calibration of each elementary intervention is separated from evaluation of their compositions.

Support: accurate sequence prediction with compatible low-level witnesses or calibrated outcome distributions, including correct rejection of unsupported projected paths. **Challenge:** isolated-step accuracy coexists with sequence failure; apparent success requires an undeclared reset; or a matched history-aware baseline performs equally well with less structure.

A failure can show that a chart is insufficient for planning while leaving its descriptive usefulness intact. The hypothesis under test must name which function the chart claims to support.

12.8 Prediction family P8: Separation and interaction of descriptive families

If ACO-derived regime descriptors or SMC-derived competence measures are included, they are independently operationalized and compared with sensory, contextual, and reporting baselines. The model predicts which variables affect immediate judgments, delayed descriptions, or transfer, without declaring every competence change a qualitative change.

Support: a prespecified incremental effect or dissociation that survives information-matched controls and independent replication. **Challenge:** the measures only restate the target reports, fail reliability, or supply no incremental prediction. This family tests the proposed application of those source frameworks; it does not validate their entire ontologies.

12.9 Why no single positive result is decisive

These prediction families overlap with established contextual, relational, or dynamical research programs. This is intentional: the proposal should enter a comparative scientific landscape rather than claim exclusive ownership of contextuality. Its incremental value must arise from a jointly constrained architecture that predicts more with fewer unsupported additions.

A model that is merely renamed after successful fitting has not tested the theory. The binding graph, target contrasts, permissible model updates, and decisive holdouts must be fixed before confirmatory analysis.

13. A Prospective Mint Research Program

13.1 Stage A: Formal and synthetic preparation

Build the record types, dimension registry, graph validation, train-test separation, uncertainty handling, and claim restrictions. Test them on synthetic data where the generating process is fully known. Synthetic trials verify implementation behavior and reveal failure modes. They supply no empirical evidence for human phenomenology.

The companion technical brief contains a small executable algebraic fixture. It demonstrates role sensitivity, contextual dependence, a commutation check, and rejection of an infeasible target. Its outputs are consequences of an explicitly chosen mathematical construction, not discovered properties of mint. A second finite-transition fixture tests full-path witnesses, cumulative budgets, incomplete-model uncertainty, and existential viability. Both are formal preparation, not psychophysical experiments.

13.2 Stage B: Within-subject descriptive calibration

Establish whether participants can reliably make the distinctions the proposed atlas requires. Use repeated comparisons and bounded rating tasks rather than assuming that a long adjective list is a valid coordinate system. Record vocabulary comprehension, response uncertainty, fatigue, and missingness.

A participant's inability to rate an investigator-selected axis should not be converted to zero intensity. It can indicate an unsuitable instrument, absent concept, uncertain report, or inapplicable dimension.

Sample size is to be determined by a preregistered precision or power analysis informed by pilot reliability. This paper does not supply an empirical effect size or assert that an arbitrary participant count is sufficient.

13.3 Stage C: Bounded factorial and compositional study

An illustrative design space comprises two aroma preparations, three independently documented delivery-strength levels, two approved temperature conditions, and two framing conditions. This yields $2 \times 3 \times 2 \times 2 = 24$ cells before repetitions and controls. It is a design example, not an exposure protocol or a claim that the factors are perceptually independent.

The delivered stimulus must be characterized independently of the report. Use neutral or balanced framing; any deception requires appropriate review and debriefing. Randomize or counterbalance order within a design that accounts

for carryover. A predefined subset of combinations remains unavailable during model fitting and selection.

Fresh-versus-dried preparations should enter a separate, explicitly matched block because they change more than one physical property. Decay-associated qualities should initially be represented only through safe, approved substitutes or nonexposure tasks; spoiled material is outside the proposed tasting protocol.

13.4 Stage D: Neural correspondence

Use an appropriate institutional collaboration or suitable existing data. A recording modality is chosen for a specific question, not because it offers the most visually compelling activity map. Define time alignment, preprocessing, nuisance controls, measurement resolution, and independent regions or feature-selection rules in advance.

Test whether binding-sensitive predictors explain held-out neural or behavioral relations beyond stimulus identity, intensity, task, motor response, and history-aware alternatives. Evidence at this stage is primarily correlational unless the design supports causal inference.

13.5 Stage E: Intervention and replication

Begin causal investigation with authorized sensory and contextual manipulations. Any direct neural intervention belongs only within independently approved professional research or clinical protocols; no stimulation settings or do-it-yourself procedure are specified here.

Distinguish the total effect of a stimulus manipulation from a hypothesized neural mediation pathway. Conditioning on a measured neural mediator can change the estimand and introduce bias; mediation claims require additional assumptions beyond randomized stimulus assignment.

An independent replication should freeze the confirmatory model and test family. Failure should be analyzed before adding dimensions or special cases. A changed ontology belongs to a subsequent study, with a new holdout.

13.6 Stage F: Sequential contrasts and continuation-sensitive charts

After elementary interventions have been independently characterized, evaluate a frozen set of action sequences. Preserve the starting-state uncertainty, actual intertrial intervals, any resets, and the full recorded intervention order. Compare the content-only chart, a history-augmented chart, and a strong dynamical baseline with equivalent information access.

A useful diagnostic is whether two conditions matched on the current response instrument nevertheless require different continuation models. A positive finding would support state refinement for that task, not prove that the initial experiences were phenomenally identical or different. The study should also include

sequences deliberately outside the model's identified domain, for which calibrated nonprediction is the appropriate response.

No extra exposure or neural intervention is authorized by this stage description. It changes the analysis question and the protocol design obligations, not the independently approved intervention boundary.

13.7 Full realization funnel for the research program

The research program becomes more precise when its design space is separated into the complete realization funnel:

Stage	Application to the Locus of Qualia program
Possibility	Generate candidate qualitative distinctions, binding organizations, histories, atlases, physical substrates, measurement channels, interventions, and countermodels, including brief, harmful, negative, incoherent, or empirically empty possibilities.
Specification	Declare H-S, H-R, and H-P; episode and observation types; dimensions; roles; contexts; charts; transforms; bridge contracts; path quantifiers; baselines; evidence kinds; permissions; and falsifiers.
CM1	Enumerate configurations that satisfy the specification and basic admissibility constraints, including expensive, invasive, fragile, low-information, or otherwise undesirable realizations.
CM2	Prefer the ethically acceptable and scientifically informative region: strong alternatives, identifiable interventions, low burden, privacy, preregistration, independent holdouts, explicit uncertainty, reversibility, and high falsifiability.
Instance Configuration	Freeze the exact participants or synthetic states, stimuli, histories, instruments, dimensions, models, splits, seeds, code, resource bounds, permissions, intervention set, bridge, path query, and claim profile.

Stage	Application to the Locus of Qualia program
Implementation Realization	Execute the selected synthetic fixture, measurement protocol, model comparison, atlas construction, bridge test, path-lifting analysis, and authorized intervention.
Evidence	Preserve outputs, negative controls, residuals, ablations, uncertainty, calibration, transport, failures, replications, observer limits, and the separate dispositions of H-S, H-R, and H-P.
Revision	Create a successor when sources, dimensions, measures, models, bridges, permissions, evidence, or interpretation change; retain failed, superseded, and conflicting records.

CM1 is therefore not the preferred study. CM2 is not the executed study. Instance Configuration is the frozen slice actually selected for Implementation Realization. The distinction prevents a general architecture from receiving evidence that belongs only to one model, dataset, instrument, or intervention regime.

14. Analysis, Baselines, and Falsification

14.1 Required baseline family

The comparison set should include a stimulus-only model; a contextual additive model; a contextual model with selected interactions; a history-aware dynamical model; a neural predictor when such measurements are available; an untyped or role-scrambled counterpart of the binding architecture; and a matched-capacity flexible model.

The role-scrambling ablation must preserve as much nonrelational information as possible. Otherwise its failure can be explained by information destruction rather than the necessity of binding roles. Context and history ablations similarly require careful interpretation: withholding predictive inputs is not by itself evidence of a uniquely semantic mechanism.

The most convincing result would be improved transfer under comparable information access, parameter budget, and model-selection effort. A modest binding model that only defeats a deliberately impoverished baseline is insufficient.

14.2 Evaluation levels

Primary outcomes should be preregistered proper predictive scores or other metrics matched to the response scale. Secondary outcomes can include calibration, rank preservation, trajectory error, relation-sensitive contrast accuracy, and intervention-commutation error. They should not be compressed into a single *consciousness score*.

Repeated observations from one participant are not independent participants. Evaluation should distinguish within-person held-out sessions, cross-context generalization, and cross-person generalization. Confidence intervals or resampling must respect participant and session dependence.

Feature extraction, normalization, dimensionality selection, graph selection, hyperparameter tuning, and chart alignment must all be learned from training data or separate calibration data. Holding out only the final regression stage does not prevent leakage.

14.3 Practical significance and equivalence

A statistically detectable improvement can be scientifically negligible. Specify a smallest relevant gain and reliability-aware tolerances before confirmatory testing. Failure to reject a difference is not proof of equivalence. A selective-control claim requires evidence of a meaningful target effect and sufficiently small off-target effects, not merely nonsignificant secondary comparisons.

Uncertainty should propagate from trial measurement through latent-coordinate estimation to intervention and transfer claims. Missing, censored, and inapplicable observations require different handling.

14.4 Falsification and revision

The following outcomes would count against particular versions of the program: no reproducible descriptive structure; binding roles providing no benefit over matched models; context effects confined to response labeling; poor held-out composition; failure of hypothesized intervention correspondence; unstable coordinates across repeat measurement; or a need for unbounded special cases.

Not every negative result falsifies every relational view of experience. Conversely, that logical fact must not be used to immunize the current implementation. The correct disposition can be *this model failed in this jurisdiction*, *this bridge is unsupported*, or *the available measurement cannot discriminate the alternatives*.

14.5 Operationalizing SBQ without turning it into a consciousness scale

The following profile is a proposed application of SBQ's five interacting dimensions [M11, Section 3], not an inherited validated instrument.

- **SBQ dimension:** Structural coherence
  Candidate research-model diagnostic: Rate of incompatible typed claims and violated declared relationships
  Required limitation: Internal consistency does not establish empirical truth
- **SBQ dimension:** Generativity
  Candidate research-model diagnostic: Correct held-out combinations or sequences per declared modeling budget
  Required limitation: Count valid predictions, not merely generated hypotheses
- **SBQ dimension:** Perturbation stability
  Candidate research-model diagnostic: Preservation of prespecified performance and calibration under bounded nuisance changes
  Required limitation: Correct detection of a real change must not be penalized as instability
- **SBQ dimension:** Constraint alignment
  Candidate research-model diagnostic: Compliance with identity, data-role, intervention, and claim-scope contracts
  Required limitation: Administrative compliance does not establish the mechanism
- **SBQ dimension:** Minimal distortion
  Candidate research-model diagnostic: Information loss in report normalization and reliance on post hoc exceptions
  Required limitation: A simpler model is not better when it erases genuine distinctions

The profile belongs to an account or transformation, not directly to the pleasantness or intensity of an experience. A well-grounded model of an aversive event can have high representational integrity. A polished but poorly supported model of a pleasant event can have low integrity. Likewise, stability of a mistaken category is not automatically a scientific virtue.

These diagnostics should remain a profile with declared denominators, uncertainty, and tradeoffs. An aggregate score, if used for a bounded engineering comparison, must not be renamed a degree of consciousness. SBQ's energy landscape is an analogy in its source, not a physical potential governing neurons [M11, Section 4].

14.6 Separate empirical findings from continuity and admission status

SMT can audit a declared transformation of research records, including a probabilistic preservation condition with explicit observation rules [M9, Sections 8-11]. The empirical evaluation must independently report whether the data support, contradict, or fail to resolve the relevant hypothesis. A required invariant that is

unobserved may block an audit claim while leaving the empirical proposition unresolved.

For example, a chart migration can lack authorized lineage and therefore fail continuity admission even when its numerical predictions match. Conversely, it can preserve every declared record field while propagating an empirically wrong model. The workbench records both results. It does not use the word *collapse* to imply phenomenal loss when only the research artifact's continuity failed.

CSC adds a separate candidate-to-admitted governance event [M14, Sections 4-5]. A precisely serialized model, an admitted research artifact, an empirically supported predictor, and a warranted phenomenal-constitution claim are four different statuses. The approval of a protocol permits its bounded execution; it cannot determine its outcome.

15. Worked Formal Illustrations

15.1 Context, roles, and coupled outputs

Consider a deliberately artificial high-level state (a, b, h), where a and b are role-distinguished components and h is a contextual or historical variable. Define

$$q = (a, b, ab + 0.25h). \tag{28}$$

This q is a synthetic descriptor, not measured phenomenology. Encode the same state at a lower level as

$$z = (a + b, a - b, h). \tag{29}$$

The inverse map recovers $a = (z_1 + z_2)/2$, $b = (z_1 - z_2)/2$, and $h = z_3$. Increasing a by δ corresponds to adding $(\delta, \delta, 0)$ to z. Applying that lower-level change and decoding produces the same result as applying the high-level transformation and recomputing (28). This is an exact commutation property by construction.

At $(a, b, h) = (0.2, 0.8, 0)$, the descriptor is $(0.2, 0.8, 0.16)$. Increasing a by 0.1 gives $(0.3, 0.8, 0.24)$. The interaction coordinate changes by 0.08 even though b remains fixed. Separate inputs do not imply orthogonal output changes.

Indeed, with h fixed, the pullback of an output Euclidean metric has the form

$$J^\top J = \begin{bmatrix} 1 + b^2 & ab \\ ab & 1 + a^2 \end{bmatrix}. \tag{30}$$

The cross term is nonzero when $ab \neq 0$. A target $(0.3, 0.8, 0.16)$ is not reachable while keeping $h = 0$ and obeying (28). It should be identified as incompatible with the construction rather than presented as a feasible independently controlled profile.

This example is intentionally simple enough that every result can be inspected. It shows what an implementation must distinguish: roles, coupling, context, transformation correspondence, and feasible targets. The relation between such a construction and actual experience remains an empirical question.

15.2 Same displayed state, different available future

The second illustration is the four-state construction in Section 9.7. Its displayed middle descriptor q_1 hides the distinction between b and c. This example makes a further kind of binding relevant: the binding between a current descriptor, the concrete state that realizes it, and the next permitted action. It is not enough to bind the descriptor to *some* state on each occasion; the selected witnesses must belong to one compatible history.

The companion fixture returns no complete lift for the two-step sequence, returns a witness after adding a compatible second transition, and requires an explicit additional step when a bridge between the two middle states is introduced. It also distinguishes a complete toy graph's negative result from an unresolved query in a partial graph. These are properties of the stated model.

Together the two illustrations show why an experiential research architecture needs more than feature arithmetic. The first checks roles and contextual coupling within a state; the second checks compatibility across a trajectory. Neither asserts that a conscious episode has already been realized.

15.3 Exact companion-fixture replay

The current exact companion source reviewed for this pass is `Qualia_Binding_Realization_Technical_Brief.md`, SHA-256 `6693AEFCFF0E46A3E2B12B9B024C971A3B973508EAFCCABCAD89C70E35612FF0`. Its two embedded Python blocks have code hashes

`6e4a47ec6329e56b250d4795f314a8e47eacc93d25598ff803b851dda5d0baf9` and `656aabd44629bf48e0fde5a35d9c1ee215215c8dd9f2c9e4fdcaebf6caaaf2c2`.

This pass independently extracted and ran those exact blocks under CPython 3.12.10 using only the standard library in a temporary directory. Fixture A returned `PASS` for eleven grouped checks; Fixture B returned

`PASS` for twenty-six; the combined count was thirty-seven. The replay record is `work/paper-saturation-v1/evidence/the-locus-of-qualia/embedded-fixture-replay.v0.1.json`, SHA-256 `CAB6DDA6BD4379117880B8AEF2F18303593F71D366648C357B4A120137F75E7F`.

This confirms only that the exact embedded algorithms execute under the stated local runtime and return their stipulated results. It does not establish

empirical fit, neural realization, phenomenal constitution, consciousness, source authenticity, or evidence readiness. The source-reported exhaustive comparison over 512 directed three-state graphs and its 2,048 generated path queries were not independently replayed because their cross-check implementation is not embedded in the reviewed artifact.

Contextual Experiential-Binding Hypothesis

Across the completed P1 corpus, The Locus of Qualia should occupy one precise role: it is the **contextual experiential-binding hypothesis**. It proposes that qualitative character may depend on bounded, structured relations among state, context, history, roles, and realization rather than on an isolated property. It does not identify a quale with its model, report, neural correlate, coordinate, or binding record.

- **Adjacent work:** *The Locus of Meaning*
  Supplies: A relational-constitutional theory of semantic binding and locus.
  Locus-of-qualia boundary: Semantic binding motivates the architecture but does not prove that qualitative character follows the same law.
- **Adjacent work:** *Qualia Binding and Realization Mapping*
  Supplies: Measurement, bridge, trial, provenance, and realization contracts.
  Locus-of-qualia boundary: That work operationalizes the research program; this paper states the experiential hypothesis and limits.
- **Adjacent work:** *The Locus of Shared Experience*
  Supplies: Difference-preserving comparison and interaction across distinct realizations.
  Locus-of-qualia boundary: Shared experience presupposes local experiential targets without merging them.
- **Adjacent work:** *Admissibility Theory of Consciousness*
  Supplies: A graded structural program for self-situated admissibility organization.
  Locus-of-qualia boundary: Admissibility regime and qualitative content are different coordinate families.
- **Adjacent work:** *Mainstream Consciousness Discourse as Lexical Disambiguation Failure*
  Supplies: Separation of overloaded targets in consciousness discourse.
  Locus-of-qualia boundary: Every result here must state whether it concerns report, content, access, realization, or phenomenality.

A qualitative research claim should bind:

$$Q = (subject, context, history, physical_state, \\ measured_state, report, content_hypothesis, \\ binding_model, realization_model, evidence, standing)$$

The fields support a claim ladder rather than a single verdict. A protocol may establish a report difference, a discriminative state, a structural correlate, a causal intervention effect, or a bounded realization bridge. None alone establishes phenomenal identity, absence, universality, or substrate independence. Operational equivalence remains weaker than phenomenal identity.

The hypothesis should be evaluated against simpler rivals: independent feature coding, report bias, task strategy, learned lexical association, measurement leakage, common-input correlation, and unmodeled temporal state. Local charts and partial bridges are admissible outcomes; failure to construct a universal manifold is not itself falsification. A positive result becomes relevant only when it discriminates among rival structures under a declared measurement model.

The universal GSM can type
`HAS_CONTEXTUAL_EXPERIENCE_REPORT`, `CONDITIONED_BY_HISTORY`,
`MEASURED_AS`, `HYPOTHESIZED_CONTENT`, `BOUND_UNDER_EXPERIENTIAL_MODEL`,
`REALIZED_BY_CANDIDATE`, `CAUSALLY_PERTURBED_BY`, `OPERATIONALLY_EQUIVALENT_TO`,
`PHENOMENALLY_UNRESOLVED_FROM`,
and `DEFEATED_BY_RIVAL`. This preserves uncertainty instead of converting a research representation into an ontological conclusion.

This integration establishes four editorial commitments: adopt contextual experiential-binding hypothesis as the official corpus role; adopt the qualitative-claim envelope; publish an explicit claim ladder from report through realization bridge; and require discrimination from simpler rival explanations before any strong qualitative inference.

Independent Admission of Experiential Bridges

The fixed-seed comparison with Water/MCIR, MDSP, *Legitimacy Without Self-Ratification*, and *Recursive Constitutional Motifs* sharpens how a qualitative claim may advance from report or measurement toward a realization bridge:

$$B = (hypothesis, subject, measurement, bridge, alternatives, \\ control_graph, adjudicator, evidence, standing, \\ remainder, revision)$$

The laboratory, model, instrument, analysis, ontology, and adjudicator may share a control domain. A bridge should not gain standing solely because the same system defined the experiential target, produced the measurements, fitted the mapping, and declared success. Independence is threat-model-relative, and

subject-originated reports remain admissible under a bounded ceiling rather than being either excluded or treated as self-proving.

MDSP could support interoperable claim and evidence packages, versioned methods, and comparison suites. It must not standardize one qualia ontology, binding model, or realization bridge into truth before the research earns that standing. Adoption, implementation, conformance, replication, and phenomenal interpretation remain distinct.

The Water and recursive-motif adjacencies do not supply scientific evidence. Water produces
`NO_MATERIAL_RELATION`; flow terms do not explain qualitative character. A recursive motif may offer a

`PEDAGOGICAL_ONLY` decomposition of hypothesis, gate, realization, and evidence, but fit to the motif neither supports the hypothesis nor establishes recursion in experience.

The universal GSM can type
`PROPOSES_EXPERIENTIAL_BRIDGE`,
`SHARES_SCIENTIFIC_CONTROL_DOMAIN`, `ADJUDICATES_BRIDGE_STANDING`,
`STANDARDIZES_EVIDENCE_PACKAGE_WITHOUT_ONTOLOGY`, `HAS_NO_MATERIAL_RELATION_TO_WATER_FLOW`,
and
`USES_MOTIF_AS_PEDAGOGY_ONLY`.

This randomized review establishes four editorial commitments: adopt the independent bridge-admission record; require control-graph disclosure for strong realization claims; limit standards to interoperable evidence and method packages; and enforce Water and motif non-evidence boundaries.

Phenomenal Bridge Claims Remain Independently Admitted

Eight downstream uses position this work beside ATC, domain standards, anti-self-ratification, qualia mapping, recursive motifs, meaning, shared experience, and MCIR projections. They confirm the need for an explicit bridge ledger: semantic, computational, behavioral, neural, and physical correspondences may constrain an experiential model without establishing qualitative identity.

$$\begin{aligned} Q^* = (&subject, target_experience, descriptor, context,\\ &history, report, behavior, physical_state, measurement,\\ &model, intervention, bridge, alternatives, uncertainty,\\ &ethics, standing, revision) \end{aligned}.$$

Every bridge must state direction, resolution, invariances, loss, subject scope, counterfactual prediction, and evidence class. Within-subject reproducibility does not imply cross-person identity; operational equivalence does not imply

phenomenal identity; neural or electronic controllability does not provide a reverse compiler to experience; and a local chart need not glue into a global manifold.

Reports are evidence shaped by access, vocabulary, demand, memory, and incentives. Non-report is not non-experience. Structural and causal findings can survive while the phenomenal interpretation fails. Ethical safeguards govern research conduct independently of whether a bridge claim is ultimately supported.

The universal GSM can type
`PROPOSES_PHENOMENAL_BRIDGE`,
`MEASURES_EXPERIENTIAL_DESCRIPTOR`, `INTERVENES_ON_PHYSICAL_SUPPORT`,
`FAILS_CROSS_SUBJECT_IDENTIFICATION`, `RETAINS_REPRESENTATIONAL_RESIDUAL`,
`WITHHOLDS_PHENOMENAL_IDENTITY`, and
`REVISES_QUALIA_MODEL`.

This review establishes four editorial commitments: adopt the bridge ledger; require directional, resolution- and subject-bound correspondence claims; preserve report, behavior, physical support, model, and experience as separate targets; and keep ethical protection independent from positive phenomenal inference.

16. Objections and Limits

16.1 Is this merely a more elaborate description?

It could be. A graph with many contextual annotations can redescribe known results without predicting anything. The decisive difference would be transferable transformation rules, reduced out-of-sample error, and independently tested intervention structure. Without those, the contribution remains a representational proposal.

16.2 Does semantic binding presuppose a conscious interpreter?

Not by the definition used here. Typed differentiation and consequence can be modeled without assuming a linguistic interpreter. However, avoiding that assumption does not establish consciousness in the modeled system. The explanatory challenge cannot be solved by defining every useful relation as already experiential.

16.3 Does context merely relocate the mystery?

A contextual constitution may explain why one experience differs from another while leaving open why any such event is experienced. That would be genuine but limited progress. The proposal is valuable only if it clarifies and constrains what needs explanation; adding the word *binding* to the unexplained phenomenon would not suffice.

16.4 Can a rich enough matrix already represent the proposal?

Potentially. A matrix or dynamical model with appropriately defined states can encode role structure, history, and context. The present hypothesis is not an argument against mathematics or computation. Its challenge is to a particular explanatory decomposition, not to numerical representation as such.

If a competitor's matrix captures the same independently tested causal organization more parsimoniously, the binding account should acknowledge that equivalence or advantage. An architectonic representation need not be the unique correct notation.

16.5 Are qualitative dimensions truly compositional?

That is an empirical question. Some qualities may combine nonlinearly, require joint description, or appear only in particular contexts. The framework permits such interactions but pays an explanatory cost for each additional dependency. Noncompositional residuals must not be hidden by assigning a new primitive to every observation.

16.6 Can structural correspondence distinguish inverted or absent experience?

Not necessarily. Observational and interventional agreement on the chosen test set may leave several phenomenal interpretations open. The identifiability limitation is not a reason to abandon experiments; it is a reason to state what their results actually discriminate.

Evidence that a binding variable plays a causal role in reporting does not alone prove that it is the felt quality. Conversely, the absence of a universally accepted phenomenal test does not negate the reality or importance of ordinary first-person reports.

16.7 Does nonfundamental spacetime follow?

No. The formalism does not derive spacetime, abolish physical objects, or establish that consciousness is ontologically prior to physics. Hoffman's observer-first research program motivates one route into the problem, but its stronger metaphysical claims are not premises of this paper. The supplied interview itself acknowledges an outstanding burden to recover relevant physical structures [T1, approximately 1:22:25-1:23:37].

The manuscript makes no claims about telepathy, anomalous aerial phenomena, altered-state entities, or access to a hidden reality. None follows from local charts, binding graphs, or Markov descriptions.

16.8 Does the source integration settle the philosophical question?

No. ACO provides one explicitly non-phenomenal boundary for its use of *consciousness*, Locus supplies a theory of semantic constitution, and H-P asks an additional question about qualitative specificity. Their joint inclusion does not constitute three independent arguments establishing the same conclusion. The disagreement over the role and measurability of qualia must remain visible [M1, Section 14.5; M6, Section 10].

Nor does the ability to describe an episode with an elaborate binding record establish that the record identifies its sufficient physical support. The proposed integration is valuable only if its distinctions lead to better measurements, predictions, or explanatory exclusions.

16.9 Does a substrate-neutral specification imply substrate-neutral experience?

No. A specification can be implemented in different media while preserving selected operations. Whether phenomenal character also transfers is a further hypothesis. Biological features, dynamical properties, intrinsic causal organization, or other constraints could matter in ways the chosen specification misses.

A deterministic simulation can validate an analysis pipeline without possessing the properties its variables describe. A simulated cooling variable is not thereby a cold sensation.

17. Ethical and Engineering Boundaries

Research with human participants requires independent ethical oversight, informed consent, appropriate sensory-exposure review, accessibility, and an explicit plan for privacy and withdrawal. No potentially spoiled stimulus is justified by the conceptual usefulness of a decay contrast. No invasive intervention is justified by a desired coordinate change.

Experience-related data can contain sensitive bodily, autobiographical, and affective information. The default technical design should minimize collection, separate identities from trial data, preserve consent restrictions, and prevent secondary use outside the declared scope.

For artificial systems, the architecture must keep semantic competence, causal organization, agency, phenomenal status, and capacity for welfare-relevant experience as distinct questions. Uncertainty should not be converted into either an unsupported declaration of sentience or an unsupported universal denial. This paper supplies no threshold for assigning rights or declaring that an intervention causes suffering. ACO's structural-harm concept concerns unauthorized binding imposition and does not require phenomenal suffering [M6, Section 7]. That is the source's normative definition, not an empirical identification of pain. The present protocol retains ordinary independently reviewed participant protections and does not infer welfare or legal status from a regime coordinate.

An evidence record can certify what was measured, what transformation was applied, and which analysis was run. It cannot certify direct possession of another system's experience merely by being complete or cryptographically intact.

18. Discussion: What a Successful Program Would Add

The most productive outcome would not be a database row declaring that mint equals a million particular numbers. It would be a family of calibrated, context-bounded models that predict how qualitative distinctions change when their physical and relational support changes.

Such a program could clarify the difference between stable identity and changing character: how an experience remains mint-like while becoming stronger, less fresh, spatially displaced, or temporally extended. It could distinguish transformations of component magnitude from transformations of binding role. It could identify when apparently independent dimensions are actually coupled, when a correspondence fails across people or contexts, and when the same displayed quality profile conceals a continuation-relevant state difference. The latter result would distinguish a descriptive atlas from a sufficient state representation for intervention planning.

The architecture also makes explanatory overreach easier to detect. A lexical name would no longer masquerade as an experiential primitive. A measured activity pattern would no longer silently become a full physical mechanism. A latent embedding would no longer silently become a universal phenomenal metric. An intervention on a sensory input would no longer silently become proof of neural mediation. A reproducible simulation would no longer silently become a conscious realization.

These distinctions do not diminish the initiating idea. They give it sharper edges. A hypothesis can encounter evidence only where it is specific enough to be contradicted.

Mathematical Description Is Not Phenomenal Capture

Public orientation

The Locus of Qualia asks what a rigorous description of a taste such as mint would need to preserve: intensity, quality, context, time, comparison, preparation, physiology, language, and subject-indexed report. Its mathematics organizes a research program. It does not capture what mint tastes like merely by assigning coordinates.

Adversarial tests

1. **Coordinate-arbitrariness attack.** Axes reflect investigator choices, vocabulary, or model geometry. Coordinates need operational definitions, calibration, and sensitivity analysis.

2. **Metric-as-phenomenology attack.** Distance in a representation is treated as experienced similarity. Behavioral, linguistic, neural, and phenomenal relations must remain distinct.
3. **Mint-overfit attack.** A framework tuned to one worked domain fails for other modalities, cultures, preparations, or unfamiliar experiences. Cross-domain tests are required.
4. **Report-bias attack.** Language, expectation, memory, demand, and available categories shape reports. Report is evidence, not transparent access.
5. **Cross-subject identity attack.** Aligned coordinates are interpreted as identical qualia. Correspondence under a measurement profile does not establish phenomenal identity.
6. **Temporal-collapse attack.** Onset, adaptation, aftertaste, expectation, and memory are compressed into one point. The object of comparison must be time-indexed.
7. **Physiology-omission attack.** Preparation and semantics are modeled while receptor, neural, bodily, or clinical differences are ignored. Scope and omitted mechanisms must be explicit.
8. **Precision-theater attack.** More symbols and decimal values imply more knowledge than the evidence supports. Uncertainty and identifiability must accompany every estimate.

Claim and evidence ceiling

The paper establishes a typed descriptive and experimental scaffold. Limited formal results hold inside declared definitions and assumptions. They do not prove a phenomenal manifold, recover subjective character, establish neural mechanism, or permit comparison of moral standing.

Empirical publication claims require current sensory-science, neuroscience, psychophysics, philosophy, and statistical sources; participant protocols; calibration; uncertainty; preregistration where appropriate; and independent replication. Author-provided references must be checked for exact support. No external research was added in this pass.

Reader-facing consequence

The public value is not a formula for taste. It is a disciplined way to state what a comparison preserves, what evidence supports it, which subject and moment it concerns, and what remains inaccessible.

19. Publication Standing and Evidence Limits

This text is an AI-assisted integrated successor edition. The paper's theory remains a proposed, falsifiable research architecture. The two exact embedded companion fixtures now have bounded local replay evidence, but no participant,

neural, causal-intervention, cross-person, artificial-consciousness, or phenomenal-constitution study has been completed here. Corpus integration improves explanatory and custody structure; it is not additional empirical corroboration.

20. Conclusion

The taste of mint may be better investigated as a family of contextually realized binding organizations than as a single intrinsically mint-like object. Locus distinguishes identity from support; architectonics organizes the relevant relations; dimensionally explicit representation makes differences and transformations inspectable. Together they suggest a route from a qualitative contrast to a calibrated model and, potentially, to an independently tested account of physical realization.

The route is conditional. Existing semantic-binding commitments do not establish phenomenal constitution. A useful experiential atlas does not establish global geometric truth. Successful neural prediction does not establish causation, and causal correspondence does not automatically establish experience. The research program must earn each bridge separately.

Its central empirical demand is clear: specify a bounded binding architecture, predict transformations and complete sequences it has not already seen, compare it with strong alternatives, and retain its failures. Locus and GTK help specify the target; Dynatology clarifies reachable change; ACO and SMC pose separate regime and competence questions; SMT and SBQ discipline the research account; and the crystallization frameworks preserve the difference between a candidate, an admitted artifact, and a supported finding. Its strongest philosophical hypothesis is equally clear: qualitative specificity may reside in the realized organization of a contextual encounter, rather than in any component taken alone.

Mint would then be neither merely the molecule nor merely its name. It would be a particular organized occurrence whose differences can, in principle, become answerable to experiment.

Appendix A. Notation and Claim Types

- **Symbol or term:** e
  Meaning: Indexed experiential episode
  Boundary: Indexing does not specify its full character
- **Symbol or term:** ι_e
  Meaning: Identity address
  Boundary: No semantic or phenomenal payload
- **Symbol or term:** Q_e
  Meaning: Qualitative event under investigation
  Boundary: Not directly equated with a report vector

- **Symbol or term:** $\mathcal{B}_e$
  Meaning: Hypothesized realized binding organization
  Boundary: Its phenomenal role remains a hypothesis
- **Symbol or term:** $\widehat{\mathcal{A}}_e$
  Meaning: Estimated architectonic description
  Boundary: Model, not automatically mechanism
- **Symbol or term:** G_e
  Meaning: Typed binding hypergraph
  Boundary: Representation of proposed dependencies
- **Symbol or term:** q_e
  Meaning: Local coordinate representation
  Boundary: Chart-relative; potentially latent
- **Symbol or term:** $\widehat{q}_e$
  Meaning: Estimated coordinate representation
  Boundary: Uncertainty and measurement model required
- **Symbol or term:** z_t
  Meaning: Physical state at a declared scale
  Boundary: Not necessarily fully observed
- **Symbol or term:** x_t
  Meaning: Instrument measurement
  Boundary: Not interchangeable with z_t
- **Symbol or term:** y_t
  Meaning: Experience-related observation
  Boundary: Not direct access to another experience
- **Symbol or term:** J
  Meaning: Experimental jurisdiction
  Boundary: Population, task, context, and scope
- **Symbol or term:** $\mathcal{R}_J$
  Meaning: Proposed realization relation
  Boundary: Not necessarily one-to-one or functional
- **Symbol or term:** ρ
  Meaning: Low-to-high abstraction map
  Boundary: Requires independent validation
- **Symbol or term:** ω
  Meaning: Intervention correspondence
  Boundary: Not guaranteed to exist for every intervention
- **Symbol or term:** H-S / H-R / H-P
  Meaning: Structural / realization / phenomenal hypotheses
  Boundary: No automatic promotion between them

Appendix B. A Minimal Claim Ledger

Each substantive result should identify its target, evidence type, assumptions, and ceiling. For example:

```
Claim: A context-sensitive model predicts intensity reports on
held-out sessions.
Evidence: Human psychophysical observations under the registered
protocol.
Ceiling: Predictive association in the specified population and task.
Not established: Unique neural realization; phenomenal identity; AI
consciousness.

Claim: A low-level perturbation and a high-level transformation commute.
Evidence: An intervention test with a frozen abstraction map.
Ceiling: Causal correspondence for the tested variables and
intervention family.
Not established: Exhaustive experience specification; universal
substrate neutrality.

Claim: The synthetic example obeys its defining equations.
Evidence: Algebra and executed software assertions.
Ceiling: Internal consistency of a stipulated construction.
Not established: Any property of human mint experience.
```

Appendix C. Origin and Source Boundaries

The motivating interview is the supplied file `Pasted text.txt`, approximately 1 hour 24 minutes in its timestamped form. Its original publication title, interviewer surname, and publication date were not independently established for this manuscript. The name Sophia is used in the transcript but is not expanded here. The transcript's bytes were available during preparation; its ordinary SHA-256 digest is:

```
49b6903f0cfc541cac1d5cabd7412a073d7ada47ae4dc529e641a8db4fdd0b50
```

This identifies the supplied text file, not the authenticity of the recording or the correctness of its claims. The discussion uses the transcript as an attributed motivating source and does not reproduce its lengthy speculative passages.

The first draft consulted author manuscripts through the document library. This revision additionally used the fourteen directly supplied files listed in the source register [M1-M2, M4-M15]. Their exact internal titles, relevant sections, and ordinary byte digests are recorded; no publication, peer-review, or canonical-release status is inferred from their presence. The identity/support distinction and the source-specific meanings of binding, consciousness, governance, and crystallization remain explicit rather than being harmonized by renaming.

Appendix D. Companion Document

The current exact companion reviewed for this pass is `Qualia_Binding_Realization_Technical_Brief.md`. It translates this research proposal into data contracts, a staged implementation design, model-comparison requirements, acceptance gates, refusal conditions, and two executable synthetic fixtures. Earlier prose named an

`_Updated` file, but this candidate does not infer that the two filenames have identical work or artifact identity. The brief is a technical specification, not a completed platform. Its embedded fixtures have no empirical connection to mint unless a later study supplies one.

Appendix E. Source Discrepancies and Explicit Local Adaptations

The source files have not been rewritten. The following decisions govern only this paper and its companion research specification.

E1. Historical definitions remain attributed. The original Locus paper uses Meta-Binding Principle for its general thesis, while the expanded paper distinguishes Semantic Binding, Recursive Meta-Binding, and Reflexive Meta-Binding. This revision uses the expanded distinction and cites the original as lineage [M1, Section 4; M2, Sections 5-6 and 10]. ACO's non-semantic meaning-binding and qualia boundary remain its own stated positions, not definitions covertly replaced by this paper [M6, Sections 2 and 10].

E2. GTK conjunctive requirements. GTK Section 6.2 describes removing an invariant as tightening the floor and adding one as loosening it. Under the conjunctive interpretation of an invariant floor used by this research implementation, the direction is reversed. Define

$$\mathrm{Adm}(I) = \{x : \forall p \in I,\ p(x) = 1\}.$$

Then $I \subseteq I'$ implies $\mathrm{Adm}(I') \subseteq \mathrm{Adm}(I)$. Adding requirements can shrink or preserve the admitted set; removing them can enlarge or preserve it. This elementary derivative rule is used locally, with the discrepancy recorded, rather than reproducing the source passage as executable logic [M5, Sections 3.1 and 6.2]. No automatic semantic-neighbor classification follows from removing a requirement.

E3. Finite existential viability. Dynatology's infinite-path definition and deletion fixed point are retained. Its separate reference to closed strongly connected components is not used as the implementation criterion [M7, Section 28]. A singleton terminal component is closed but has no infinite path; a cyclic component with an outgoing edge can still support an infinite path. The correct finite existential criterion used here is reachability to a cycle-containing component, including a singleton only when it has a self-loop. The companion

implementation uses sink deletion, which directly realizes the source's fixed-point definition.

E4. Directed costs are not automatically symmetric metrics. The paper uses only finite nonnegative edge weights and their directed reachability cost. It does not import stronger metric, symmetry, trajectory-equivalence, or topological-closure claims from other Dynatology sections. Zero-cost edges can make distinct states have zero path cost. This has no automatic implication for qualitative indistinguishability.

E5. Empirical uncertainty does not rewrite SMT. The main SMT source's continuity decision is registrational and authority-relative. This research application records that audit separately from an empirical disposition. Missing permitted evidence can block a continuity claim without demonstrating a physical or phenomenal violation. The worked example's broader SMT/GMB language remains pedagogical rather than a replacement for the main source definition [M9, Sections 9-11; M10, "Advanced Thesis"].

E6. Historical names are not silently rewritten. The direct author ruling establishes the terminology lineage `CCF -> CMB -> GMB`, with Governed Meaning Base (GMB) current. The SBQ notes' earlier CCF-to-CMB/CX instruction and unresolved Verbatim Overlay Discipline status remain part of their historical source record [M12]. Supplied SBQ, Meaning as Binding, and CBT artifacts retain their exact CCF/CMB wording. This candidate uses GMB in new current-architecture synthesis while preserving those titles, quotations, and historically scoped descriptions [M13, Abstract; Sections 2.6 and 6].

E7. Crystallization names different operations. CSC is an admission-boundary theory in the supplied manuscript. It is not treated as an interchangeable name for SemCrys controlled-transformation test generation. Meaning Crystals supplies a bounded representational metaphor; its language does not identify physical crystals, neural energy, or an onset of phenomenal experience. The research artifacts can implement normalization and evidence tracking without claiming a deployed MCIR, GMB, or AFS integration.

Appendix F. Revision Record and Source Custody

This is an integrated successor draft, not an amendment of the fourteen source manuscripts. The original academic paper and technical brief remain separate preserved inputs. The revision adds source-specific responsibility assignments, typed identity relations, content/regime/competence/evidence separation, complete-path realization, bounded viability, SBQ-derived model diagnostics, and explicit empirical-versus-audit dispositions. It retains the original measurement limits, strong comparison models, intervention tests, and H-S/H-R/H-P separation.

The path counterexample, endpoint recurrence, and implementation contracts are derivative constructions of this revision. They are not represented as results already proved experimentally in the source corpus. Reported implementation

checks concern synthetic fixtures only. No CCOO result, prior deployment claim, or other source-reported experiment is imported as empirical evidence for mint or consciousness.

The following register contains ordinary SHA-256 digests of the supplied source files and the two parent drafts. These identify bytes for comparison; they do not confer authenticity, truth, or constitutional admission. The digest register excludes this document and its companion to avoid self-reference.

```
M1: The Locus of Meaning.md
  sha256:
  0c112b06562be7ce75652ee97cf9eaac8ee9a9486fe820e8038f5d0cd4479701
M2: What Locus of Meaning Adds.md
  sha256:
  ac11630ea96dbfd08238c6717ad3cc5eade57c98c3a4d50acc52d9e71d5e148c
M4: Meaning_as_Binding.md
  sha256:
  a9ea6370e77a71a997f1d1cac18ef6213125c6f3e13beebc9f5e8bfb94fe0b45
M5: 07_GTK_with_Atoms_merger.md
  sha256:
  4daddb53c80ab79435e3b634f11b64e70a319991248169b665c18c5aae03bb1d
M6: Ableman_Consciousness_Ontology.md
  sha256:
  fc81c7ffa89cb3aaea83b258baa1abe7aae34a1944b44a3e5d620c35632494e0
M7: Dynatology.md
  sha256:
  8539bbb8d636c406800a38c7ec47be36b1eeebfa04fae062ff6e53137099befc
M8: Semantic Manifold Competence.md
  sha256:
  4741b9908eb54864badc91f58e2a308d799b7e5f80f82c6bb0c22758c8e4cdc4
M9: Semantic Manifold Topology.md
  sha256:
  098d34f3989b4f7a01f394e99383c1faa439e39c4b951b9a50078baab6d9c314
M10: Semantic Manifold Topology - Worked Example.md
  sha256:
  baa41d5fcc820805f23676eed9b1542e0dc8744f7d7a145d21f4be99a0d99bc9
M11: Semantic_Binding_Quality.md
  sha256:
  a4f2ffc0c05684d01abcfe03503c41329517057d0001a653e2d76765d483a9ba
M12: 06_SBQ_revision_notes.md
  sha256:
  b85943479bfdff745d22bd7ff1806db34ebc63fb4caae221bc7510c84df57c7b
M13: Constitutional Binding Theory.md
  sha256:
  ad0ef1999f2cd00bfd4128e31d3ce755f746bb7160423c041921875fff1d2ddc
M14: 04_Constitutional_Semantic_Crystallization.md
  sha256:
```

```
  a7d4f1fc0254119a9ea9eb969635c91ef6bb0cae060affaf2ca87ee040d6c2f2
M15: Meaning_Crystals.md
  sha256:
  09169048cbcd0faa2baeb72c25c9bb1d51480c16c7392ee6dee51aa7e33368aa
PARENT_PAPER: The_Locus_of_Qualia_Academic_Paper.md
  sha256:
  6b0c89237d37671cd5cc741610f8225d4254d904e6b03aa25487658eb589fcc0
PARENT_BRIEF: Qualia_Binding_Realization_Technical_Brief.md
  sha256:
  138b5bc25b4b514f90dfcd3d70f00da290177510d954c7e9b15e05db69499377
```

Editorial Provenance and Standing

Source and Preparation Standing

This AI-assisted integrated edition is derived from the exact September 17, 2026 source draft identified in Appendix F. The initiating hypothesis, conceptual direction, and Locus/Meta-Binding lineage remain attributable to Adam Ableman. The source itself records AI-assisted drafting, formalization, and literature checking. This pass adds AI-prepared editorial integration, evidence qualification, and author-decision surfaces; those additions do not acquire author standing unless adopted. The source remains unchanged. No external research was performed in this pass, and the retained literature register was not independently revalidated. No human experiment, neural recording study, artificial-consciousness demonstration, GMB implementation, or phenomenal-constitution result is asserted.

Corpus-Wide Reference Positioning

This paper argues that what would it mean to specify the taste of mint mathematically?. Its nearest external comparison fields are consciousness, semantic. The sources in [E1], [E2], [E3], [E4], [E5], [E6], [E7] are cited as predecessor literature, adjacent standards, or counterpressure - not as evidence that they caused the Ableman framework or that they validate its novel claims. The comparison isolates where established work already supplies provenance, formal assurance, institutional constraint, empirical method, or engineering practice, and where this paper adds a distinct claim about admissibility, authority, semantic custody, or governed realization.

Within the Ableman corpus, the highest-priority current cross-links are [A1], [A2], [A3], [A4], [A5], [A6], [A7]. These are links among separately custodied works, not silent identity mergers. Their role is to expose mutual dependencies, terminology collisions, and places where a reader should traverse the corpus before treating this paper as self-sufficient.

Reference-pass counterevidence and limit search

The pass applied the following pressure tests: Structural and computational correspondences remain insufficient as phenomenal evidence unless linked to discriminating empirical predictions against competing theories. Formal vocabularies and graph links make commitments explicit but cannot by themselves establish correct identity, shared interpretation, or lossless translation between contexts. These limits are not generic disclaimers: they identify the assumptions that a later empirical, formal, or implementation review must try to break.

Reference-pass influence and novelty boundary

No newly added external source is asserted to be a documented authorial influence unless the predecessor source already said so. `Predecessor` means earlier relevant work;

`adjacent` means a useful comparison; `standard` means an interoperability or assurance baseline; and `counterevidence` marks work that narrows or pressures a claim. Novelty therefore remains a reviewable, claim-level question rather than an inference from a low similarity score or an absent citation.

References

External scholarship, standards, and prior art

[**E1**] David J. Chalmers. "Facing Up to the Problem of Consciousness." *Journal of Consciousness Studies* 2(3), 1995, pp. 200-219. https://consc.net/papers/facing.html **Relationship:** Foundational framing of the explanatory gap; cited to bound, not resolve, phenomenal claims.

[**E2**] Larissa Albantakis et al. "Integrated Information Theory (IIT) 4.0: Formulating the Properties of Phenomenal Existence in Physical Terms." *PLOS Computational Biology* 19(10), 2023, e1011465. DOI: 10.1371/journal.pcbi.1011465. https://doi.org/10.1371/journal.pcbi.1011465 **Relationship:** Competing formal theory of consciousness included for explicit contrast and counterevidence obligations.

[**E3**] Stanislas Dehaene and Jean-Pierre Changeux. "Experimental and Theoretical Approaches to Conscious Processing." *Neuron* 70(2), 2011, pp. 200-227. DOI: 10.1016/j.neuron.2011.03.018. https://doi.org/10.1016/j.neuron.2011.03.018 **Relationship:** Competing global-workspace account included to prevent a corpus-native ontology from substituting for empirical neuroscience.

[**E4**] Nikolaus Kriegeskorte, Marieke Mur, and Peter A. Bandettini. "Representational Similarity Analysis - Connecting the Branches of Systems Neuroscience." *Frontiers in Systems Neuroscience* 2, 2008, 4. DOI: 10.3389/neuro.06.004.2008. https://doi.org/10.3389/neuro.06.004.2008 **Relationship:** Predecessor empirical method for comparing representational geometries.

[**E5**] Francesco Locatello et al. "Challenging Common Assumptions in the Unsupervised Learning of Disentangled Representations." *PMLR* 97, 2019, pp. 4114-4124. https://proceedings.mlr.press/v97/locatello19a.html **Relationship:** Counterevidence showing that desired latent factorization is not identifiable from unsupervised data without inductive biases.

[**E6**] Richard Cyganiak, David Wood, and Markus Lanthaler, eds. *RDF 1.1 Concepts and Abstract Syntax*. W3C Recommendation, 25 February 2014. https://www.w3.org/TR/rdf11-concepts/ **Relationship:** Adjacent graph data model and identity baseline.

[**E7**] Alistair Miles and Sean Bechhofer, eds. *SKOS Simple Knowledge Organization System Reference*. W3C Recommendation, 18 August 2009. https://www.w3.org/TR/skos-reference **Relationship:** Adjacent standard for concepts, labels, schemes, notes, and semantic mappings.

Source-head external references retained under original labels

Author-provided conceptual and motivating sources

[**M12**] *Semantic Binding Quality - Revision Notes: Reviewer's edit guide for the existing draft*. Undated supplied editorial document; reviewer identity not stated. File:
`06_SBQ_revision_notes.md`. Relevant: required terminology edits, unresolved overlay term, explicit pairing with Meaning as Binding, and the statement that this is a revision guide rather than the paper. Not an independent research result.

[**T1**] Hoffman, D. D., in conversation with an interviewer addressed as Sophia. Undated interview transcript supplied by the author as `Pasted text.txt`. Relevant passages: approximately 19:59-24:30, the demand for a specified quality-to-structure account; 28:36-33:08, observer windows and recursive policies; 1:22:25-1:23:37, outstanding theoretical obligations. Original publication metadata not verified.

Corpus integration sources introduced by this pass

[**R1**] Chalmers, D. J. (1995). Facing up to the problem of consciousness. *Journal of Consciousness Studies, 2*(3), 200-219. Author-hosted text consulted.

[**R2**] Gardenfors, P. (2000). *Conceptual Spaces: The Geometry of Thought*. MIT Press. Original publication 2000; publisher's paperback listing is dated 2004. ISBN 9780262572194 identifies that paperback edition.

[**R3**] Kriegeskorte, N., Mur, M., & Bandettini, P. A. (2008). Representational similarity analysis - connecting the branches of systems neuroscience. *Frontiers in Systems Neuroscience, 2*, 4. doi:10.3389/neuro.06.004.2008.

[**R4**] Fink, S. B., Kob, L., & Lyre, H. (2021). A structural constraint on neural correlates of consciousness. *Philosophy and the Mind Sciences, 2*. doi:10.33735/phimisci.2021.79. Bibliographic details and research context checked against author-maintained material; the publisher's full text was not accessible during preparation.

[**R5**] Howard, J. D., Plailly, J., Grueschow, M., Haynes, J.-D., & Gottfried, J. A. (2009). Odor quality coding and categorization in human posterior piriform cortex. *Nature Neuroscience, 12*(7), 932-938. doi:10.1038/nn.2324. Indexed primary abstract consulted.

[**R6**] McKemy, D. D., Neuhausser, W. M., & Julius, D. (2002). Identification of a cold receptor reveals a general role for TRP channels in thermosensation. *Nature, 416*, 52-58. doi:10.1038/nature719. Indexed primary abstract and bibliographic record consulted.

[**R7**] de Araujo, I. E., Rolls, E. T., Velazco, M. I., Margot, C., & Cayeux, I. (2005). Cognitive modulation of olfactory processing. *Neuron, 46*(4), 671-679. doi:10.1016/j.neuron.2005.04.021. Primary abstract and publisher-indexed description consulted; detailed full-text methods were not relied upon.

[**R8**] Lee, B. K., et al. (2023). A principal odor map unifies diverse tasks in olfactory perception. *Science, 381*(6661), 999-1006. doi:10.1126/science.ade4401. Indexed primary abstract consulted.

[**R9**] Albantakis, L., et al. (2023). Integrated information theory (IIT) 4.0: Formulating the properties of phenomenal existence in physical terms. *PLOS Computational Biology, 19*(10), e1011465. doi:10.1371/journal.pcbi.1011465.

[**R10**] Hoffman, D. D., & Prakash, C. (2014). Objects of consciousness. *Frontiers in Psychology, 5*, 577. doi:10.3389/fpsyg.2014.00577.

Ableman corpus

[**A1**] Adam Ableman. *Ableman Consciousness Ontology (ACO)*. Ableman Corpus current edition; work identity `PAPER.PUB.18203749`;
source SHA-256
`ACD97032780DA7B3A36750702BE1F47BCB2E6361EF6E74ECA8B441D7C6A25DDE`.
Corpus relationship: priority cross-link selected by corpus profile and similarity review; no identity merger or publication effect.

[**A2**] Adam Ableman. *The Phenomenal Aperture*. Ableman Corpus current edition; work identity
`AMCS-LOCAL-WORK-4B275CBB7EFF1A4F`;
source SHA-256
`B6AE95D72B89DAF47778BED3982843A1ECE350CA363750108AA60D5A4C63D763`.
Corpus relationship: priority cross-link selected by corpus profile and similarity review; no identity merger or publication effect.

[**A3**] Adam Ableman. *Possible Minds: Against the Provincialization of Consciousness.* Ableman Corpus current edition; work identity `PAPER.PUB.20681869`;
source SHA-256
`77F381A29E2EF801085938DB369F3F432321CCEB8B9DD3969330337C24BA4998`.
Corpus relationship: priority cross-link selected by corpus profile and similarity review; no identity merger or publication effect.

[**A4**] Adam Ableman. *Admissibility Theory of Consciousness.* Ableman Corpus current edition; work identity
`AMCS-LOCAL-WORK-8FDC994841DA6CF3`;
source SHA-256
`F409652555CEC4898DBEC1E8BBB179CEE020E7FF35F4C8F307CDA82253D6F49F`.
Corpus relationship: priority cross-link selected by corpus profile and similarity review; no identity merger or publication effect.

[**A5**] Adam Ableman. *Meaning as Binding.* Ableman Corpus current edition; work identity
`AMCS-LOCAL-WORK-5F976917F0B283EA`;
source SHA-256
`A470D1A0B9ED7C3CEAB6BB88BE4B642A2BA36427447F787764329FD574DA0C97`.
Corpus relationship: priority cross-link selected by corpus profile and similarity review; no identity merger or publication effect.

[**A6**] Adam Ableman. *Governed Meaning Base (GMB).* Ableman Corpus current edition; work identity
`AMCS-LOCAL-WORK-D95CCC4FB7B58BFB`;
source SHA-256
`A3B62E50F8BEE125FDAE0D38C2839AFD1CCC138296268BF93217481E38AA48EA`.
Corpus relationship: priority cross-link selected by corpus profile and similarity review; no identity merger or publication effect.

[**A7**] Adam Ableman. *Semantic Manifold Topology.* Ableman Corpus current edition; work identity `PAPER.PUB.20257244`;
source SHA-256
`B8F28CC652A73A3EE4CF6FD0E23C1ABC922A6660319B2FB4CC1CCAAA606EC64A`.
Corpus relationship: priority cross-link selected by corpus profile and similarity review; no identity merger or publication effect.

Ableman Corpus references [**M1**] Adam Ableman (undated draft consulted September 17, 2026). *The Locus of Meaning: The Meta-Binding Principle and a Formal Architecture for Semantic Constitution.* Author's Ableman Corpus edition, stored as
`The Locus of Meaning.md`. Relevant sections: 6, "The Location of Meaning"; 7, "Semantic Binding Atlases, Manifolds, and Lattices"; 13, "Semantic Governance"; 14.5, "No automatic theory of consciousness."

[**M2**] Adam Ableman (undated draft consulted September 17, 2026). *The Locus*

of Meaning: The Semantic Binding Principle, Recursive Meta-Binding, and a Formal Architecture for Semantic Constitution. Author's Ableman Corpus edition stored under the filename
`What Locus of Meaning Adds.md`. The internal title differs from the filename. Relevant material: Abstract and opening definitions.

[M3] Adam Ableman (undated technical draft consulted September 17, 2026). *Technical Brief: Semantic Field Mechanics as the Foundational Layer of Hyperpedia.* Stored as
`Semantic_Field_Mechanics_Technical_Brief.md`. The source labels itself an experimental candidate. Relevant sections: "Transformation-First Hypothesis," "Semantic Locus Constraint," "ESF Role," and "Mathematical Machinery."

[M4] Adam Ableman (undated supplied manuscript). *Meaning as Binding: A Relational Ontology for Constitutional Computational Substrates.* File:
`Meaning_as_Binding.md`. Relevant: Sections 3-4, relational binding and meaning-inertia; Section 8, oscillatory interpretation; Section 9, binding quality. Historical CCF terminology is retained in the source.

[M5] Adam Ableman (supplied draft labeled Version 0.1; date placeholder unresolved). *Global Term Key: Identity Primitives in Governed Knowledge Systems.* Subtitle: *Disambiguated Semantic Binding as the Lexical Lattice for Constitutional Reasoning.* File:
`07_GTK_with_Atoms_merger.md`. Relevant: Sections 1-4, binding identity and lexical constitution; Section 6.2, amendment wording qualified in Appendix E.

[M6] Adam Ableman (undated supplied manuscript). *Ableman Consciousness Ontology (ACO): Consciousness as a Manifold of Meaning-Binding, Situational Awareness, and Epistemic Self-Authorship.* File:
`Ableman_Consciousness_Ontology.md`. Relevant: Sections 2-5, constraint relevance, historical S-to-C-to-I grammar, ESA, and regime axes; Section 7, structural harm; Section 10, qualia boundary.

[M7] Adam Ableman (undated supplied manuscript). *Dynatology: A Science of Structured Potentiality.* File: `Dynatology.md`. Relevant: Sections 3, 6, 8, 11-12, 25, and 28. Only the bounded transition, reachability, and existential-viability constructions are used here. Source-reported CCOO experiments are not qualia evidence and are not independently revalidated by this revision.

[M8] Adam Ableman (undated supplied manuscript). *Semantic Manifold Competence: Reinterpreting Intelligence, Cognition, and the Turing Test Through Structured Meaning Topologies.* File:
`Semantic Manifold Competence.md`. Relevant: Section 2, scope and five capacities; Sections 6.2-6.3, candidate memory and attention interpretations; Section 18.3, the surrounding framework stack.

[M9] Adam Ableman (undated supplied manuscript). *Semantic Manifold Topology: Constitutional Reachability, Collapse, and Governance-Relative Continuity.* File:

`Semantic Manifold Topology.md`. Relevant: Abstract; Sections 3, 8-11, and 21. This is the diagnostic SMT definition used by the present revision.

[**M10**] Adam Ableman (undated supplied worked example). *Semantic Manifold Topology: A Simple Worked Example: Conversation.* File:
`Semantic Manifold Topology - Worked Example.md`. Relevant: thick origin-binding, confidence versus admissibility, and the "Advanced Thesis" passage. Used as exposition with its scope difference from [M9] preserved.

[**M11**] Adam Ableman (undated supplied manuscript). *Semantic Binding Quality (SBQ): A Structural Framework for Evaluating Meaning Stability.* File:
`Semantic_Binding_Quality.md`. Relevant: Sections 2-4, presence, interacting quality dimensions, and energy analogy; Section 8, candidate computational indicators. Source terminology remains unamended.

[**M13**] Adam Ableman (undated supplied manuscript). *Constitutional Binding Theory: Governed Anchor Resolution as a Universal Pattern in Semantic Systems.* File:
`Constitutional Binding Theory.md`. Relevant: Sections 1.1-1.5, evidence and amendment; Section 9, premature collapse and fragmentation; historical CMB/CCF wording remains source-specific.

[**M14**] Adam Ableman (supplied draft labeled Version 0.1; date placeholder unresolved). *Constitutional Semantic Crystallization: The Phase Transition by Which Meaning Acquires Binding Authority.* File:
`04_Constitutional_Semantic_Crystallization.md`. Relevant: Sections 4-5, canonical/admitted distinction, nonthermodynamic metaphor, and correctness boundary.

[**M15**] Adam Ableman (undated supplied manuscript). *Meaning Crystals: Language, MCIR, and the Physics of Legible Thought.* File: `Meaning_Crystals.md`. Relevant: Sections 1-2, representation and stabilization; Section 5, oracle surfaces; Section 11, limits of the crystal analogy.

[**C1**] Adam Ableman *Semantic Legitimacy Before Context.* Current reviewed head:
`source-material/`

`Semantic Legitimacy Before Context.md`; SHA-256
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[**C3**] Adam Ableman *The Three Hash Rule.* Current reviewed head:
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[**C4**] Adam Ableman *A Formal Theory of Work Custody.* Current reviewed head:
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[**C6**] Adam Ableman *Semantic Debt.* Current reviewed head:
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`36BAE578CF1926981488982AB8C406E0EACB662E25E8B5F18F187A6E53AF933F`.

[**C7**] Adam Ableman *Legitimacy Without Self-Ratification.* Current reviewed head:
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These records were used as internal corpus interfaces through the verified Ableman corpus skill. They were not treated as independent scientific evidence for the qualitative-binding hypothesis, and their publication, version-family, and identity standings remain separately governed.

Published research and primary scholarly sources

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Corpus source and companion formats: https://github.com/AblemanMosaic/ableman-corpus
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